

**15.1 Projection of a side of a Triangle**

Understand the following theorems along with their corollaries and their applications to solve the allied problems.

**Theorem 15.1.1**

In an obtuse-angled triangle, the square on the side opposite to the obtuse angle is equal to the sum of the squares on the sides containing the obtuse angle together with twice the rectangle contained by one of the sides, and the projection on it of the other.

**Given:**

$\triangle ABC$  with an obtuse angle at vertex A.

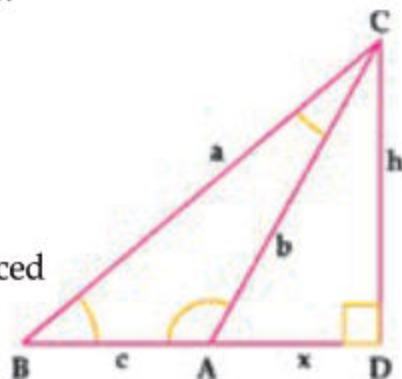
**To prove:**

i.e.  $a^2 = b^2 + c^2 + 2cx$

**Construction:**

Draw perpendicular  $\overline{CD}$  on  $\overline{BA}$  produced meeting at point D, so that  $\overline{AD}$  is the projection of  $\overline{AC}$  on  $\overline{BA}$  produced.

Taking,  $m\overline{BC} = a$ ,  $m\overline{CA} = b$ , and  $m\overline{AB} = c$  also  $m\overline{AD} = x$  and  $m\overline{CD} = h$ .



**Proof:**

| Statements                                                                                              | Reasons                                            |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| In right angled $\triangle CDA$<br>$m\angle CDA = 90^\circ$                                             | Construction                                       |
| so, $(m\overline{AC})^2 = (m\overline{AD})^2 + (m\overline{DC})^2$                                      | By Pythagoras theorem.                             |
| i.e. $b^2 = x^2 + h^2 \dots (i)$                                                                        | By supposition                                     |
| In right angled $\triangle CDB$<br>$m\angle CDB = 90^\circ$                                             | Construction                                       |
| so, $(m\overline{BC})^2 = (m\overline{BD})^2 + (m\overline{CD})^2$                                      | By Pythagoras theorem.                             |
| i.e. $a^2 = (c+x)^2 + h^2$                                                                              | $m\overline{BD} = m\overline{BA} + m\overline{AD}$ |
| or $a^2 = c^2 + 2cx + x^2 + h^2 \dots (ii)$                                                             | $b^2 = x^2 + h^2$                                  |
| $a^2 = c^2 + 2cx + b^2$                                                                                 |                                                    |
| Thus $(m\overline{BC})^2 = (m\overline{AB})^2 + 2(m\overline{AB})(m\overline{AD}) + (m\overline{AC})^2$ |                                                    |

**Q.E.D**



**Example:** In a  $\triangle ABC$  with obtuse angle is at vertex A, if  $\overline{CD}$  is an altitude on  $\overline{BA}$  produced, meeting at point D, and  $m\overline{AC} = m\overline{AB}$ .  
 Then,  $(m\overline{BC})^2 = 2(m\overline{AB})(m\overline{BD})$ .

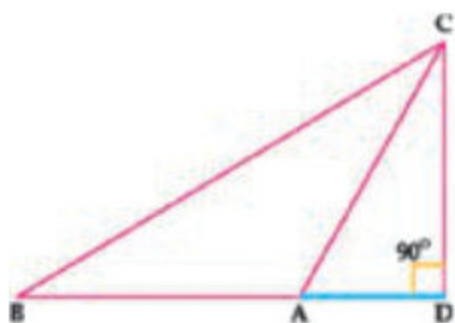
**Given:**

In a  $\triangle ABC$ ,  $m\angle A$  is an obtuse,  
 $m\overline{AC} = m\overline{AB}$  and  $\overline{CD}$  being altitude on  
 $\overline{BA}$  produced, meeting at point D.

**To prove:**

$$(m\overline{BC})^2 = 2(m\overline{AB})(m\overline{BD}).$$

**Proof:**



| Statements                                                                                         | Reasons                                            |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------|
| In a $\triangle ABC$                                                                               |                                                    |
| $(m\overline{BC})^2 = (m\overline{AB})^2 + 2(m\overline{AB})(m\overline{AD}) + (m\overline{AC})^2$ | Theorem 15.1.1                                     |
| $(m\overline{BC})^2 = (m\overline{AB})^2 + 2(m\overline{AB})(m\overline{AD}) + (m\overline{AB})^2$ | Given that $m\overline{AC} = m\overline{AB}$       |
| $(m\overline{BC})^2 = 2(m\overline{AB})^2 + 2(m\overline{AB})(m\overline{AD})$                     | Taking $2m\overline{AB}$ as common                 |
| $(m\overline{BC})^2 = 2m\overline{AB}(m\overline{AB} + m\overline{AD})$                            | $m\overline{BD} = m\overline{AD} + m\overline{AB}$ |
| $(m\overline{BC})^2 = 2(m\overline{AB})(m\overline{BD})$                                           |                                                    |

**Q.E.D**



## Exercise 15.1

- Find the length of  $\overline{AB}$  and area of the triangle  $ABC$ , when
  - $m\overline{AC} = 3\text{ cm}$ ,  $m\overline{BC} = 6\text{ cm}$  and  $m\angle C = 120^\circ$ , where  
 $m\overline{CD} = m\overline{BC} \cos(180^\circ - m\angle C)$
  - $m\overline{AC} = 40\text{ mm}$ ,  $m\overline{BC} = 80\text{ mm}$  and  $m\angle C = 120^\circ$ , where  
 $m\overline{CD} = m\overline{BC} \cos(180^\circ - m\angle C)$   
 Hint:  $(m\overline{BC})^2 = (m\overline{AC})^2 + (m\overline{AB})^2 + 2(m\overline{AB})(m\overline{AD})$
- Find the length of  $m\overline{AC}$  in the  $\triangle ABC$  when  $m\overline{BC} = 6\text{ cm}$ ,  $m\overline{AB} = 4\sqrt{2}\text{ cm}$  and  $m\angle ABC = 135^\circ$ . If possible, find the area of the  $\triangle ABC$ .
- Find the length of  $m\overline{AC}$  in the  $\triangle ABC$  when  $m\overline{BC} = 6\sqrt{2}\text{ cm}$ ,  $m\overline{AB} = 8\text{ cm}$  and  $m\angle ABC = 135^\circ$ . If possible, find the area of the  $\triangle ABC$ .

## Theorem 15.1.2

In any triangle, the square on the side opposite to an acute angle is equal to sum of the squares on the sides containing that acute angle diminished by twice the rectangle contained by one of those sides and the projection on it of the other.

**Given:**

$\triangle ABC$  with an acute  $\angle CAB$  at vertex  $A$ .

Taking,  $m\overline{BC} = a$ ,  $m\overline{AC} = b$  and  $m\overline{AB} = c$

**Construction:**

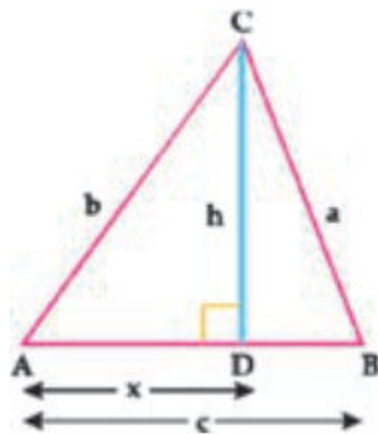
Draw  $\overline{CD} \perp \overline{AB}$  so that  $\overline{AD}$  is projection of  $\overline{AC}$  on  $\overline{AB}$

Also  $m\overline{AD} = x$  and  $m\overline{CD} = h$

**To prove:**

$$(m\overline{BC})^2 = (m\overline{AC})^2 + (m\overline{AB})^2 - 2(m\overline{AB})(m\overline{AD})$$

i.e.  $a^2 = b^2 + c^2 - 2cx$




**Proof:**

| Statements                                                                                                                                                                                                                                                                    | Reasons                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| In right angled $\triangle CDA$ ,<br>$m\angle CDA = 90^\circ$ ,<br>so, $(m\overline{AC})^2 = (m\overline{AD})^2 + (m\overline{CD})^2$<br>i.e. $b^2 = x^2 + h^2 \quad \dots (i)$                                                                                               | Construction<br>Using Pythagoras Theorem.<br>By supposition                                                                 |
| In right angled $\triangle CDB$<br>$m\angle CDB = 90^\circ$<br>$(m\overline{BC})^2 = (m\overline{BD})^2 + (m\overline{CD})^2$<br>so, i.e.<br>$a^2 = (c-x)^2 + h^2$<br>$a^2 = c^2 - 2cx + x^2 + h^2$<br>or $a^2 = c^2 - 2cx + b^2 \quad \dots (ii)$<br>$a^2 = c^2 + b^2 - 2cx$ | Construction<br>By Pythagoras Theorem.<br>From the figure.<br>$\therefore m\overline{BD} = m\overline{AB} - m\overline{AD}$ |
| Hence<br>$(m\overline{BC})^2 = (m\overline{AC})^2 + (m\overline{AB})^2 - 2(m\overline{AB})(m\overline{AD})$                                                                                                                                                                   | Using equation... (i)                                                                                                       |

Q.E.D

### Apollonius and the theorem of Apollonius:

Apollonius was a great geometer and astronomer.

Now we state and prove one of his well-known theorem "the Apollonius theorem".

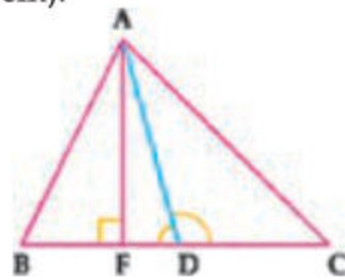
#### Theorem 15.1.3 (Apollonius theorem)

In any triangle, the sum of the squares on any two sides is equal to twice the square on half of the third side together with twice the square on the median which bisects the third side, (Apollonius' theorem).

**Given:**

In  $\triangle ABC$ , the median  $\overline{AD}$  bisects  $\overline{BC}$  at point D.

such that  $m\overline{BD} = m\overline{CD}$ .



**To prove:**

$$(m\overline{AB})^2 + (m\overline{AC})^2 = 2(m\overline{BD})^2 + 2(m\overline{AD})^2$$

**Construction:**Draw  $\overline{AF} \perp \overline{BC}$ .**Proof:**

| Statements                                                                                                          | Reasons                                                                           |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| In $\triangle ADB$<br>Since, $\angle ADB$ is acute                                                                  | $\triangle ADF$ is right angled triangle with right angled at F<br>(construction) |
| So,<br>$(m\overline{AB})^2 = (m\overline{BD})^2 + (m\overline{AD})^2 - 2(m\overline{BD})(m\overline{FD}) \dots (i)$ | Using theorem 15.1.2                                                              |
| Now, In $\triangle ADC$<br>We $\angle ADC$ is an obtuse angle at point D.                                           | Supplement of in $\angle ADB$                                                     |
| So,<br>$(m\overline{AC})^2 = (m\overline{CD})^2 + (m\overline{AD})^2 + 2(m\overline{CD})(m\overline{FD})$           | Theorem 15.1.1                                                                    |
| $(m\overline{AC})^2 = (m\overline{BD})^2 + (m\overline{AD})^2 + 2(m\overline{BD})(m\overline{FD}) \dots (ii)$       | $m\overline{CD} = m\overline{BD}$                                                 |
| Thus,<br>$(m\overline{AB})^2 + (m\overline{AC})^2 = 2(m\overline{BD})^2 + 2(m\overline{AD})^2$                      | Adding eqs. (i) and (ii)                                                          |

**Q.E.D**



### Exercise 15.2

1. In  $\triangle ABC$ ,  $m\angle A = 30^\circ$ , prove that  

$$(m\overline{BC})^2 = (m\overline{AC})^2 + (m\overline{AB})^2 - \sqrt{3}(m\overline{AB})(m\overline{AC}).$$
2. In a  $\triangle ABC$ , calculate  $m\overline{BC}$  when  $m\overline{AB} = 6\text{cm}$ ,  $m\overline{AC} = 5\text{cm}$ , and  $m\angle A = 60^\circ$
3. Whether the triangle with sides 3cm, 4cm and 5cm is acute, obtuse or right angled.
4. Find the length of the median of side  $\overline{BC}$  of a  $\triangle ABC$  where  $m\overline{AB} = 4\text{cm}$ ,  $m\overline{AC} = 3\text{cm}$  and  $m\overline{BC} = 6\text{cm}$ .

### Review Exercise 15

1. Fill in the blanks.
  - (i) In rt.  $\triangle ABC$ ,  $(m\overline{AB})^2 + \underline{\hspace{2cm}} = (m\overline{AC})^2$ ,
  - (ii) In  $\triangle ABC$ , two sides are equal to 4cm it is called \_\_\_\_\_ triangle.
  - (iii) In  $\triangle ABC$ , with  $m\angle B = 90^\circ$  then  
 $(m\overline{AB})^2 + (m\overline{BC})^2 = \underline{\hspace{2cm}}$
  - (iv) 8cm, 15cm and 17 cm are the sides of \_\_\_\_\_
2. In  $\triangle ABC$ ,  $m\overline{AC} = 3\text{cm}$ ,  $m\overline{BC} = 6\text{cm}$  and  $m\angle C = 120^\circ$ . Compute  $m\overline{AB}$



## Summary

- ◆ In an obtuse-angled triangle, the square on the side opposite to the obtuse angle is equal to the sum of the squares on the sides containing the obtuse angle together with twice the rectangle contained by one of the sides, and the projection on it of the other.
- ◆ In any triangle, the square on the side opposite to an acute angle is equal to the sum of the squares on the sides containing that acute angle diminished by twice the rectangle contained by one of those sides and the projection on it of the other.
- ◆ In any triangle, the sum of the squares on any two sides is equal to twice the square on half the third side together with twice the square on the median which bisects the third side, (Apollonius theorem).

## Unit 16

• Weightage = 5%

# Introduction To Coordinate Geometry / Analytical Geometry

### Student Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- ◆ Explain and define coordinate geometry.
- ◆ Derive distance formula to calculate distance between two points given in Cartesian plane.
- ◆ Use distance formula to find distance between two given points.
- ◆ Define collinear points. Distinguish between collinear and non- collinear points.
- ◆ Use distance formula to show that three (or more) given points are collinear.
- ◆ Use distance formula to show that the given three non-collinear points form:
  - i. An equilateral triangle,
  - ii. An isosceles triangle,
  - iii. A right angled triangle,
  - iv. A scalene triangle.
- ◆ Use distance formula to show that four given non- collinear points form:
  - ◆ A square,
  - ◆ A rectangle,
  - ◆ A parallelogram.
- ◆ Recognize the formula to find the midpoint of the line joining two given points.
- ◆ Apply distance and midpoint formulas to solve/verify different standard results related to geometry.

## Introduction

The Cartesian coordinate was invented in the 17th century by Rene Descartes (Latinized name as Cartesius) revolutionized mathematics by providing the first systematic link between Euclidean geometry and algebra. Using the Cartesian co-ordinate system, geometric shapes (such as lines and curves) can be described by equations.

### 16.1 Distance Formula

#### 16.1.1 Explain and define Coordinate Geometry

Co-ordinate geometry is one of the most important and exciting branch of mathematics. In particular it is central to the mathematics students meet at school. It provides a connection between algebra and geometry through graphs of lines and curves.

The algebraic study of geometry with the help of coordinate system is called co-ordinate geometry / analytical geometry.

This enables geometrical problems to be solved algebraically and provides geometric insights into algebra. It is a part of geometry in which ordered pairs of numbers are used to describe the position of a point on a plane. Here, the concept of coordinate geometry (also known as Cartesian geometry) and its formulas and their derivations will be explained.

#### 16.1.2 Derive distance formula to calculate the distance between two given points in the Cartesian plane

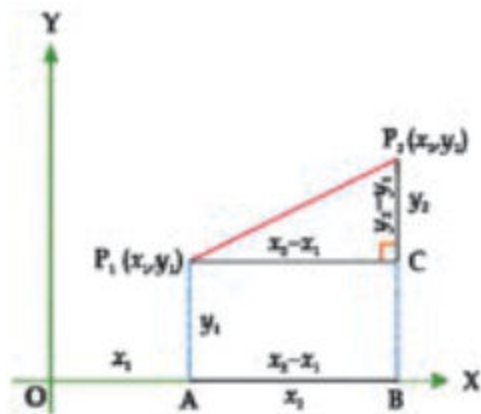
##### Statement:

The distance between any two points  $P_1(x_1, y_1)$  and  $P_2(x_2, y_2)$  is denoted as  $|P_1P_2|$  and is defined as:

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

##### Derivation of the Distance Formula

Let  $P_1(x_1, y_1)$  and  $P_2(x_2, y_2)$  be the any two points in plane.





From  $P_1$  and  $P_2$  draw perpendiculars  $\overline{P_1A}$  and  $\overline{P_2A}$  on  $x$ -axis, Also draw a  $\overline{P_1C}$  parallel to  $x$ -axes.

$$|\overline{P_1C}| = |\overline{AB}| = |\overline{OB}| - |\overline{OA}| = |x_2 - x_1|$$

$$\text{and } |\overline{P_2C}| = |\overline{P_2B}| - |\overline{BC}| = |y_2 - y_1|$$

Consider right angled  $\Delta P_1CP_2$  and Applying Pythagoras theorem, we have,

$$\therefore |\overline{P_1P_2}|^2 = |\overline{P_1C}|^2 + |\overline{P_2C}|^2$$

$$\Rightarrow |\overline{P_1P_2}|^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\Rightarrow |\overline{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**Note:**

The distance  $d$  from origin to the point  $P(x, y)$  is:

$$d = \sqrt{x^2 + y^2}$$

### 16.1.3 Use Distance Formula to find the distance between two given points.

The following examples will help to understand the use of distance formula.

**Example 01** Find the distance between the point  $P(2, 3)$  and  $Q(-4, 5)$

**Solution:** By using distance formula  $|\overline{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Here

$$(x_1, y_1) = (2, 3), (x_2, y_2) = (-4, 5)$$

$$\therefore |\overline{PQ}| = \sqrt{(-4 - 2)^2 + (5 - 3)^2}$$

$$\Rightarrow |\overline{PQ}| = \sqrt{(-6)^2 + (2)^2} = \sqrt{36 + 4}$$

$$\Rightarrow |\overline{PQ}| = \sqrt{40} = 2\sqrt{10}$$



**Example 02** Circle with radius 5 unit is drawn with centre  $C(3,2)$  and  $L(6,6)$ ,  $M(0,-1)$  and  $N(-2,-3)$  points are given. Find which of the point is not on the circle. (give reason).

**Solution:**

$C(3,2)$  is the centre of a circle with radius 5 units.

and  $L(6,6)$ ,  $M(0,-1)$   $N(-2,-3)$  are three given points.

We know that,

We have to, find the length (distance) from  $C$  to  $L$ ,  $M$  and  $N$  respectively using distance formula, then,

$$\therefore |CL| = \sqrt{(6-3)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5 \text{ units,}$$

$$|CM| = \sqrt{(0-3)^2 + (-1-2)^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \text{ units,}$$

$$|CN| = \sqrt{(-2-3)^2 + (-3-2)^2} = \sqrt{25+25} = \sqrt{50} = 5\sqrt{2} \text{ units,}$$

$$|CL| = 5 \text{ units i.e, radius of the circle}$$

$$|CM| = 3\sqrt{2} \text{ units} < 5 \text{ units}$$

$$|CN| = 5\sqrt{2} \text{ units} > 5 \text{ units}$$

So,  $M$  and  $N(-2,-3)$  are not on the circle.

### Exercise 16.1

1. Using distance formula find the distance between the following pairs of points.

(i)  $(-4,5)$  and  $(6,6)$

(ii)  $(2,2)$  and  $(2,3)$

(iii)  $(0,1)$  and  $(2,3)$

(iv)  $(0,1)$  and  $(2,3)$

2.  $A(a,0)$  and  $B(0,b)$  be the points on the axes, find the distance between  $A$  and  $B$ , when

(i)  $a = -3$ ,  $b = -4$

(ii)  $a = -9$ ,  $b = 6$

(iii)  $a = 3$ ,  $b = 4$

(iv)  $a = \sqrt{2}$ ,  $b = -2\sqrt{2}$

3. Find the perimeter of the triangle formed by the points.

$A(0,0)$ ,  $B(4,0)$  and  $C(2,2\sqrt{3})$ .



## 16.2 Collinear Points

### 16.2.1 Define collinear points. Distinguish between collinear and non-collinear points.

#### Collinear Points

##### Definition:

Three or more points are said to be collinear if they lie on the same line.

In the following figure



Points A, B and C are collinear points i.e.  $|\overline{AC}| = |\overline{AB}| + |\overline{BC}|$

#### Non-Collinear Points

##### Definition:

Three or more points are said to be non-collinear points, if they do not lie on same line.

In the following figure



Points A, B and C are non-collinear points.

##### Notes:

- Three non-collinear points form a triangle and four non-collinear points form a quadrilateral.
- If points P, Q and R are collinear, then either
  - $|\overline{PR}| = |\overline{PQ}| + |\overline{QR}|$
  - or  $|\overline{PQ}| = |\overline{PR}| + |\overline{RQ}|$
  - or  $|\overline{QR}| = |\overline{QP}| + |\overline{PR}|$  holds good, otherwise the points are non-collinear.



### 16.2.2 Use Distance Formula to show that three (or more) given points are collinear.

**Example 01** Show that the points  $A(3, -2)$ ,  $B(1, 4)$  and  $C(-3, 16)$  are collinear points.

**Solution:**

$A(3, -2)$ ,  $B(1, 4)$  and  $C(-3, 16)$  are three given points.

Now we find the distances  $|\overline{AB}|$ ,  $|\overline{BC}|$  and  $|\overline{AC}|$  by using distance formula.

$$\therefore |\overline{AB}| = \sqrt{(1-3)^2 + (4+2)^2} = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10} \text{ units,}$$

$$|\overline{BC}| = \sqrt{(-3-1)^2 + (16-4)^2} = \sqrt{16+144} = \sqrt{160} = 4\sqrt{10} \text{ units,}$$

$$|\overline{AC}| = \sqrt{(-3-3)^2 + (16+2)^2} = \sqrt{36+324} = \sqrt{360} = 6\sqrt{10} \text{ units,}$$

$$\text{Here, } |\overline{AB}| + |\overline{BC}| = 2\sqrt{10} + 4\sqrt{10} = 6\sqrt{10} = |\overline{AC}| = d_3,$$

Therefore the points  $A$ ,  $B$  and  $C$  are collinear points. Shown.

**Example 02** Using distance formula show that  $A(-2, -3)$ ,  $B(4, 7)$  and  $C(9, -5)$  non-collinear?

**Solution:**

$A(-2, -3)$ ,  $B(4, 7)$  and  $C(9, -5)$  are given three points.

Now find the distances  $|\overline{AB}|$ ,  $|\overline{BC}|$  and  $|\overline{AC}|$  by using distance formula.

$$\therefore |\overline{AB}| = \sqrt{(4+2)^2 + (7+3)^2} = \sqrt{36+100} = \sqrt{136} = 2\sqrt{34} \text{ units,}$$

$$\Rightarrow |\overline{BC}| = \sqrt{(9-4)^2 + (-5-7)^2} = \sqrt{25+144} = \sqrt{169} = 13 \text{ units,}$$

$$\Rightarrow |\overline{AC}| = \sqrt{(9+2)^2 + (-5+3)^2} = \sqrt{121+4} = \sqrt{125} = 5\sqrt{5} \text{ units,}$$

$$\text{Since, } |\overline{AC}| \neq |\overline{AB}| + |\overline{BC}|$$

Thus, the given three points  $A$ ,  $B$  and  $C$  are not collinear.



### 16.2.3 Use distance formula to show that the given three non-collinear points forms.

- (i) An equilateral triangle,
- (ii) An isosceles triangle,
- (iii) A right angled triangle,
- (iv) A scalene triangle.

**Example 01** Show that the three points  $A(1,1)$ ,  $B(-1,-1)$  and  $C(-\sqrt{3},\sqrt{3})$  form an equilateral triangle.

**Solution:**

$A(1,1)$ ,  $B(-1,-1)$  and  $C(-\sqrt{3},\sqrt{3})$  are given points.

Now, find the distance  $|\overline{AB}|$ ,  $|\overline{BC}|$  and  $|\overline{AC}|$  of the sides of a  $\triangle ABC$

using distance formula

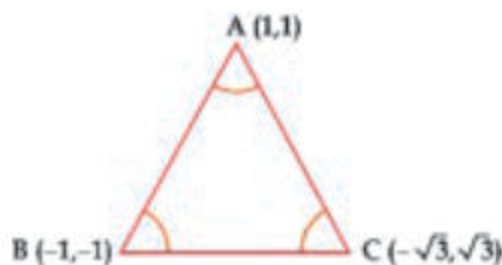
$$|\overline{AB}| = \sqrt{(-1-1)^2 + (-1-1)^2} = \sqrt{4+4} = \sqrt{8} \text{ units,}$$

$$|\overline{BC}| = \sqrt{(-\sqrt{3}+1)^2 + (\sqrt{3}+1)^2} = \sqrt{3-2\sqrt{3}+1+3+2\sqrt{3}+1} = \sqrt{8} \text{ units,}$$

$$\text{and } |\overline{AC}| = \sqrt{(-\sqrt{3}-1)^2 + (\sqrt{3}-1)^2} = \sqrt{4+4} = \sqrt{8} \text{ units,}$$

Since,  $|\overline{AB}| = |\overline{BC}| = |\overline{AC}| = 2\sqrt{2}$  units, and the points are non-collinear.

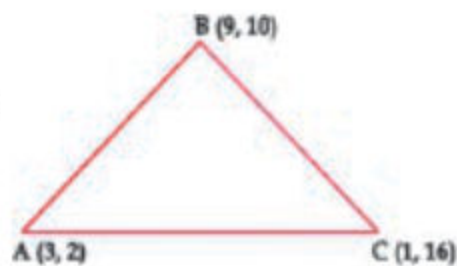
Therefore  $ABC$  is an equilateral triangle.



**Example 02** Show that the points  $A(3,2)$ ,  $B(9,10)$  and  $C(1,16)$  form an isosceles triangle.

**Solution:**

Let  $A(3,2)$ ,  $B(9,10)$  and  $C(1,16)$  are given points.



By using distance formula

$$|AB| = \sqrt{(9-3)^2 + (10-2)^2} = \sqrt{36+64} = \sqrt{100} = 10 \text{ units,}$$

$$|AC| = \sqrt{(1-3)^2 + (16-2)^2} = \sqrt{4+196} = \sqrt{200} = 10\sqrt{2} \text{ units,}$$

$$|BC| = \sqrt{(1-9)^2 + (16-10)^2} = \sqrt{64+36} = \sqrt{100} = 10 \text{ units,}$$

Since,  $|AB| = |BC| = 10$  unit and points are non-collinear.

Thus, the two sides are equal in length,

Therefore  $ABC$  is an isosceles triangle.

**Example 03** Show that  $A(2, 1)$ ,  $B(5, 1)$  and  $C(2, 6)$  are the vertices of a right angled triangle.

**Solution:**

Let  $A(2, 1)$ ,  $B(5, 1)$  and  $C(2, 6)$  be the vertices of a  $\triangle ABC$ .

Using distance formula,

$$|AB|^2 = (5-2)^2 + (1-1)^2 = 9+0=9$$

$$|BC|^2 = (2-5)^2 + (6-1)^2 = 9+25=34$$

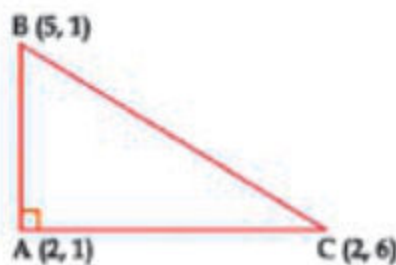
and  $|AC|^2 = (2-2)^2 + (6-1)^2 = 0+25=25$

Here,

$$|AB|^2 + |AC|^2 = 9+25=34$$

$$\Rightarrow |AB|^2 + |AC|^2 = |BC|^2$$

By converse of Pythagoras theorem given vertices form a right angled triangle.





**Example 04** Show that the point  $A(3,4)$ ,  $B(1,2)$  and  $C(0,4)$  form a scalene triangle.

**Solution:**

Let  $A(3,4)$ ,  $B(1,2)$  and  $C(0,4)$  are the given points.

Now, find the length of each side using distance formula

$$\therefore |AB| = \sqrt{(1-3)^2 + (2-4)^2} = \sqrt{4+4} = \sqrt{8} \text{ units,}$$

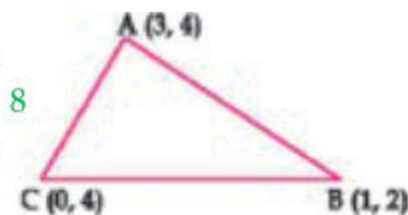
$$|BC| = \sqrt{(0-1)^2 + (4-2)^2} = \sqrt{1+4} = \sqrt{5} \text{ units,}$$

$$\text{and } |AC| = \sqrt{(0-3)^2 + (4-4)^2} = \sqrt{9} = 3 \text{ units,}$$

Since,  $|AB| \neq |BC| \neq |AC|$  and the points are non-collinear.

i.e. length of all the three sides are not equal.

Thus ABC is a scalene triangle.



### 16.2.4 Use distance formula to show that four non-collinear points form:

- A parallelogram.
- A rectangle,
- A square

**Example 01** Show that  $A(-8,-3)$ ,  $B(-2,6)$ ,  $C(8,5)$  and  $D(2,-4)$  consecutive vertices of a parallelogram.

**Solution:**

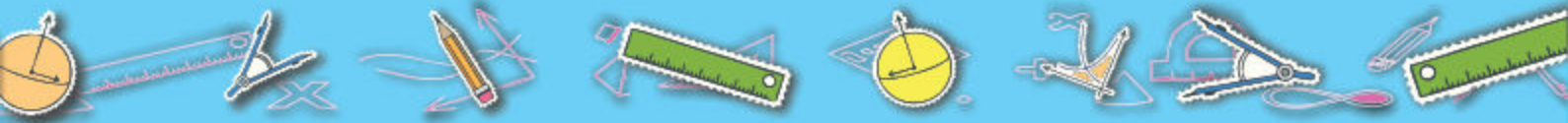
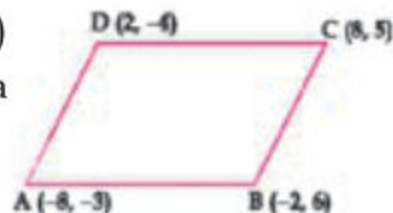
Let  $A(-8,-3)$ ,  $B(-2,6)$ ,  $C(8,5)$  and  $D(2,-4)$  be the any four consecutive vertices of a quadrilateral ABCD.

Using Distance Formula,

We have,

$$|AB| = \sqrt{(-2+8)^2 + (6+3)^2} = \sqrt{36+81} = \sqrt{117} \text{ units,}$$

$$|BC| = \sqrt{(8+2)^2 + (5-6)^2} = \sqrt{100+1} = \sqrt{101} \text{ units,}$$



$$|\overline{DC}| = \sqrt{(2-8)^2 + (-4-5)^2} = \sqrt{36+81} = \sqrt{117} \text{ units,}$$

$$\text{and } |\overline{AD}| = \sqrt{(2+8)^2 + (-4+3)^2} = \sqrt{100+1} = \sqrt{101} \text{ units,}$$

$$\text{Now, } |\overline{AB}| = |\overline{CD}| = \sqrt{117}$$

$$\text{and } |\overline{BC}| = |\overline{AD}| = \sqrt{101}$$

$\therefore$  A, B, C and D are the vertices of parallelogram.

**Example 02** Show that the four points A(0, -1), B(4, -3), C(8, 5) and D(4, 7) are the consecutive vertices of a rectangle.

**Solution:**

Let, A(0, -1), B(4, -3), C(8, 5) and D(4, 7) be any four consecutive points of a quadrilateral ABCD.

Using distance formula,

$$\text{i.e., } d = |\overline{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \text{ unit}$$

We have,

$$\therefore |\overline{AB}| = \sqrt{(4-0)^2 + (-3+1)^2} = \sqrt{16+4} = \sqrt{20} \text{ units,}$$

$$|\overline{BC}| = \sqrt{(8-4)^2 + (5+3)^2} = \sqrt{16+64} = \sqrt{80} \text{ units,}$$

$$|\overline{DC}| = \sqrt{(4-8)^2 + (7-5)^2} = \sqrt{16+4} = \sqrt{20} \text{ units,}$$

$$|\overline{AD}| = \sqrt{(4-0)^2 + (7+1)^2} = \sqrt{16+64} = \sqrt{80} \text{ units,}$$

$$|\overline{AC}| = \sqrt{(8-0)^2 + (5+1)^2} = \sqrt{64+36} = 10 \text{ units,}$$

$$|\overline{BD}| = \sqrt{(4-4)^2 + (7+3)^2} = \sqrt{0+100} = 10 \text{ units,}$$

$$\therefore |\overline{AB}| = |\overline{CD}| = \sqrt{20}$$

$$|\overline{BC}| = |\overline{AD}| = \sqrt{80}$$

$$|\overline{AC}| = |\overline{BD}| \quad (\text{Diagonals are equal})$$

A, B, C and D are the vertices of a rectangle.

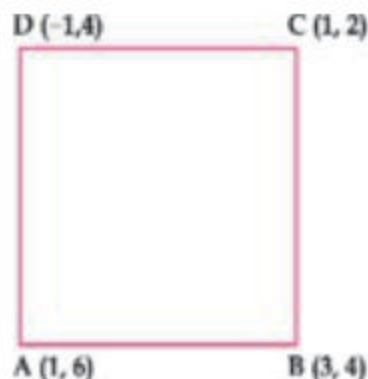




**Example 03** Show that the four consecutive points  $A(1,6)$ ,  $B(3,4)$ ,  $C(1,2)$  and  $D(-1,4)$  form a square.

**Solution:**

Given that  $A(1,6)$ ,  $B(3,4)$ ,  $C(1,2)$  and  $D(-1,4)$  be the four consecutive points of a quadrilateral ABCD.



By using the distance formula

$$\therefore |\overline{AB}| = \sqrt{(3-1)^2 + (4-6)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \text{ units,}$$

$$|\overline{BC}| = \sqrt{(1-3)^2 + (2-4)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \text{ units,}$$

$$|\overline{DC}| = \sqrt{(1+1)^2 + (2-4)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \text{ units,}$$

and  $|\overline{AD}| = \sqrt{(-1-1)^2 + (4-6)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \text{ units,}$

Since,  $|\overline{AB}| = |\overline{BC}| = |\overline{DC}| = |\overline{AD}| = 2\sqrt{2}$  unit, i.e. four sides are equal in length.

Now, find the lengths of the diagonals  $\overline{AC}$  and  $\overline{BD}$  respectively

$$\therefore |\overline{AC}| = \sqrt{(1-1)^2 + (2-6)^2} = \sqrt{0+16} = 4 \text{ units,}$$

and  $|\overline{BD}| = \sqrt{(-1-3)^2 + (4-4)^2} = \sqrt{16+0} = 4 \text{ units,}$

Since,  $|\overline{AC}| = |\overline{BD}| = 4$  unit, i.e. lengths of the diagonals are equal.

Hence, a quadrilateral ABCD is a Square.



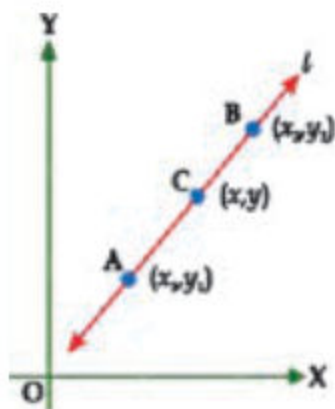
### Exercise 16.2

1. Show that the points P (-3, -4), Q (2, 6) and R (0, 2) are collinear.
2. Show that the points A (-1, 0), B (1, 0) and C (0,  $\sqrt{3}$ ) are not collinear.
3. Show that L (0,  $\sqrt{3}$ ), M (-1, 0) and N (1, 0) form an equilateral triangle.
4. Whether or not the points A (2, 3), B (8, 11) and C (0, 17) form an isosceles triangle.
5. Do the points A (-1, 2), B (7, 5) and C (2, -6) form a right angled triangle.
6. If the points A (3, 1), B (9, 1) and C (6, k) determine an equilateral triangle, find the values of k.
7. Show that the points P (1, 2), Q (3, 4) and R (0, -1) are the vertices of a scalene triangle.
8. Show that the points A (2, 3), B (8, 11), C (0, 17) and D (-6, 9) are vertices of a square.
9. Explain why the points A (-2, 0), B (0, -3), C (2, 0) and D (0, 3) do determine a square?
10. Show that the points A (3, 2), B (4, 1), C (5, 4) and D (6, 3) are the vertices of a rectangle.
11. Use distance formula to show that the points O (0, 0), A (3, 0), B (5, 2) and C (2, 2) form the vertices of a parallelogram.

## 16.3 Mid-Point Formula

### 16.3.1 Recognize the formula to find the mid-point of the line joining two given points.

Let A ( $x_1, y_1$ ) and B ( $x_2, y_2$ ) be any two point of the  $\overline{AB}$  in the plane and C ( $x, y$ ) be the midpoint of AB, then  $C(x, y) = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$  is called mid point of A and B.





**Example 01** Find the mid-point of the line segment joining A (2, 1) and B (3, 4)

**Solution:**

A (2, 1) and B (3, 4) are the points of the line segment.

Mid point = ?

Using mid-point formula

$$\therefore \text{Mid-point of } \overline{AB} = \left( \frac{2+3}{2}, \frac{1+4}{2} \right) = \left( \frac{5}{2}, \frac{5}{2} \right)$$

**Example 02** If A(2, 1), B(5, 2), and C(3, 4) are the vertices of a  $\triangle ABC$ , find the mid-points P, Q and R of the sides  $\overline{AC}$ ,  $\overline{AB}$  and  $\overline{BC}$  respectively of a  $\triangle ABC$ .

**Solution:**

Using mid-point formula,

$$C(x, y) = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \text{ we have,}$$

$$\therefore \text{Mid-point of } \overline{AC} = P = \left( \frac{2+3}{2}, \frac{1+4}{2} \right) = \left( \frac{5}{2}, \frac{5}{2} \right),$$

$$\text{Mid-point of } \overline{AB} = Q = \left( \frac{2+5}{2}, \frac{1+2}{2} \right) = \left( \frac{7}{2}, \frac{3}{2} \right),$$

$$\text{and, Mid-point of } \overline{BC} = R = \left( \frac{5+3}{2}, \frac{2+4}{2} \right) = (4, 3),$$

Thus, the required mid-points of the sides of a  $\triangle ABC$  are

$$\left( \frac{5}{2}, \frac{5}{2} \right), \left( \frac{7}{2}, \frac{3}{2} \right) \text{ and } (4, 3).$$

## 16.4 Apply Distance And Midpoint Formulas To Solve/Verify Different Standards Results Related To Geometry

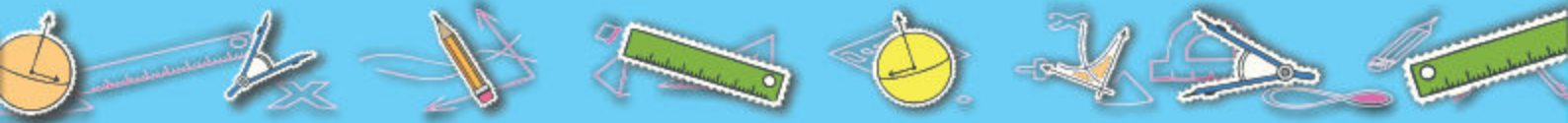
**Example 01** Prove analytically that the length of the median to the hypotenuse of a right triangle is half the length of the hypotenuse.

**Solution:**

Let ABC be the triangle right-angled at B. Take B as a origin and  $\overline{BC}$ ,  $\overline{BA}$  as the axes of x and y respectively as shown in the figure

Let  $|\overline{BC}| = a, |\overline{BA}| = b$  so that B is (0,0), C is (a,0) and A is (0,b).

Therefore M, the midpoint of  $\overline{AC}$  is  $\left( \frac{a+0}{2}, \frac{b+0}{2} \right)$ , i.e.  $\left( \frac{a}{2}, \frac{b}{2} \right)$ .



$$\begin{aligned}
 \text{Now } |\overline{AM}| &= |\overline{CM}| = \frac{1}{2} |\overline{AC}| \\
 &= \frac{1}{2} \sqrt{|\overline{AB}|^2 + |\overline{BC}|^2} \\
 &= \frac{1}{2} \sqrt{a^2 + b^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } |\overline{BM}| &= \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2} - 0\right)^2} \\
 &= \frac{1}{2} \sqrt{a^2 + b^2}
 \end{aligned}$$

Therefore

$$|\overline{BM}| = \frac{1}{2} |\overline{AC}|$$

Hence the length of median  $|\overline{BM}|$  is half the length of the hypotenuse  $|\overline{AC}|$ .

**Example 02** Prove that the figure obtained by joining in order the midpoints of the sides of any quadrilateral is parallelogram.

**Solution:**

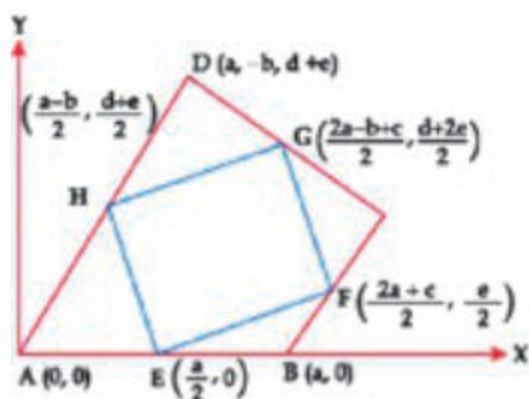
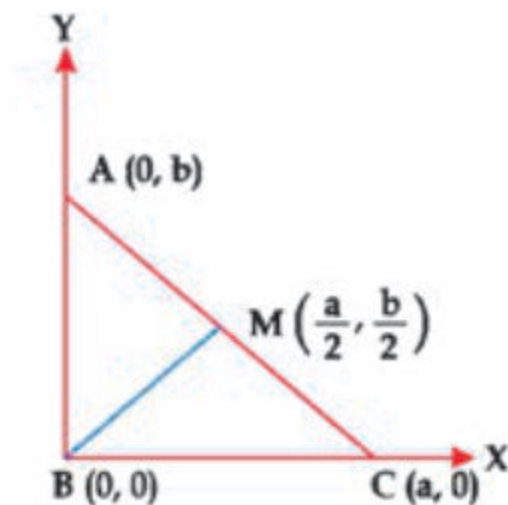
Let ABCD be a quadrilateral and E, F, G and H respectively be the midpoints of the sides  $\overline{AB}$ ,  $\overline{BC}$ ,  $\overline{CD}$  and  $\overline{DA}$ .

Let A be taken as origin,  $\overline{AB}$  as x-axis and a line through A and perpendicular to  $\overline{AB}$  as y-axis

Then A is (0,0) again let B be (a,0), C be (a+c, e) and D be (a-b, d+e)

Now the midpoints E, F, G and H are

respectively  $\left(\frac{a}{2}, 0\right)$ ,  $\left(\frac{2a+c}{2}, \frac{e}{2}\right)$ ,  $\left(\frac{2a-b+c}{2}, \frac{d+2e}{2}\right)$  and  $\left(\frac{a-b}{2}, \frac{d+e}{2}\right)$ .





Therefore, by the distance formula

$$\begin{aligned} |\overline{EF}|^2 &= \left( \frac{2a+c}{2} - \frac{a}{2} \right)^2 + \left( \frac{e}{2} - 0 \right)^2 \\ &= \left( \frac{a+c}{2} \right)^2 + \left( \frac{e}{2} \right)^2 \\ &= \frac{1}{4} \{ (a+c)^2 + e^2 \} \end{aligned} \quad \text{(i)}$$

$$\begin{aligned} |\overline{GH}|^2 &= \left( \frac{2a-b+c}{2} - \frac{a-b}{2} \right)^2 + \left( \frac{d+2e}{2} - \frac{d+e}{2} \right)^2 \\ &= \left( \frac{a+c}{2} \right)^2 + \left( \frac{e}{2} \right)^2 \\ &= \frac{1}{4} \{ (a+c)^2 + e^2 \} \end{aligned} \quad \text{(ii)}$$

From (i) and (ii), we have

$$|\overline{EF}| = |\overline{GH}|$$

Similarly,  $|\overline{GF}| = |\overline{EH}|$

Since the opposite sides are equal, EFGH is a parallelogram.

### Exercise 16.3

- Find the mid-points between the following pair of points using mid-point formula.
  - A (2, 6) and B (-4, 8)
  - P (-3, -1) and Q (5, 2)
  - L (0, 6) and M (-8, 0)
  - C (0, 0) and D ( $2\sqrt{3}, 4\sqrt{3}$ )
- Find the centre of a circle whose end points of a diameter are A (-5, 6) and B (3, -4).
- The centre of a circle is (3, 4) and one of its end point of a diameter is (4, 6), find the point of other end.
- A circle has a diameter between the points A (-3, 4) and B (11, 6). Find the centre and radius of the circle.
- Prove that a triangle is an isosceles triangle if and only if it has two equal medians.
- Prove that the diagonals of a parallelogram bisect each other.



### Review Exercise 16

1. Read the following sentences carefully and encircle "T" in case of True and "F" in case of False statement.

- (i)  $R$  is the mid-point of  $\overline{PQ}$ , if  $R$  is lying between  $P$  and  $Q$  T/F
- (ii) In scalene triangle all sides are equal T/F
- (iii) Perpendicular lines meet at an angle of  $135^\circ$ . T/F
- (iv) Collinear points may form a triangle. T/F
- (v) Non-collinear points form a triangle. T/F
- (vi) In an isosceles triangle, two sides angles are equal. T/F
- (vii) All the points that lie on the  $y$ -axis are collinear T/F
- (viii) Intersection of  $x$ -axis and  $y$ -axis is  $(0,0)$  T/F
- (ix) The distance from origin to  $(6,0)$  is 36 unit. T/F

2. Fill in the blanks.

- (i) If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  be the any two points on the line, then  $|\overline{AB}| = \underline{\hspace{2cm}}$
- (ii) Collinear points lies on the same  $\underline{\hspace{2cm}}$
- (iii) In 4<sup>th</sup> quadrant  $x > 0$  and  $y \underline{\hspace{2cm}}$

3. Tick (✓) the correct answer.

- (i) Two perpendicular lines meet at an angle of:
  - (a)  $45^\circ$  (b)  $60^\circ$
  - (c)  $90^\circ$  (d)  $180^\circ$
- (ii)  $\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$  is called:
  - (a) Mid-point formula (b) Distance formula
  - (c) Division formula (d) Ratio formula
- (iii)  $A(3, 0)$  and  $B(0, 3)$  are any two points in the plane then  $|\overline{AB}| =$ 
  - (a) 6 unit (b)  $6\sqrt{2}$  unit
  - (c)  $3\sqrt{2}$  unit (d)  $3\sqrt{2}$  unit
- (iv) Three points  $A, B$  and  $C$  are collinear if
  - (a)  $m \overline{AB} = m \overline{BC} + m \overline{AC}$  (b)  $(m \overline{AB})^2 = (m \overline{BC})^2 + (m \overline{AC})^2$
  - (c) both a and b (d)  $(m \overline{AB}) \neq m \overline{BC} + m \overline{AC}$

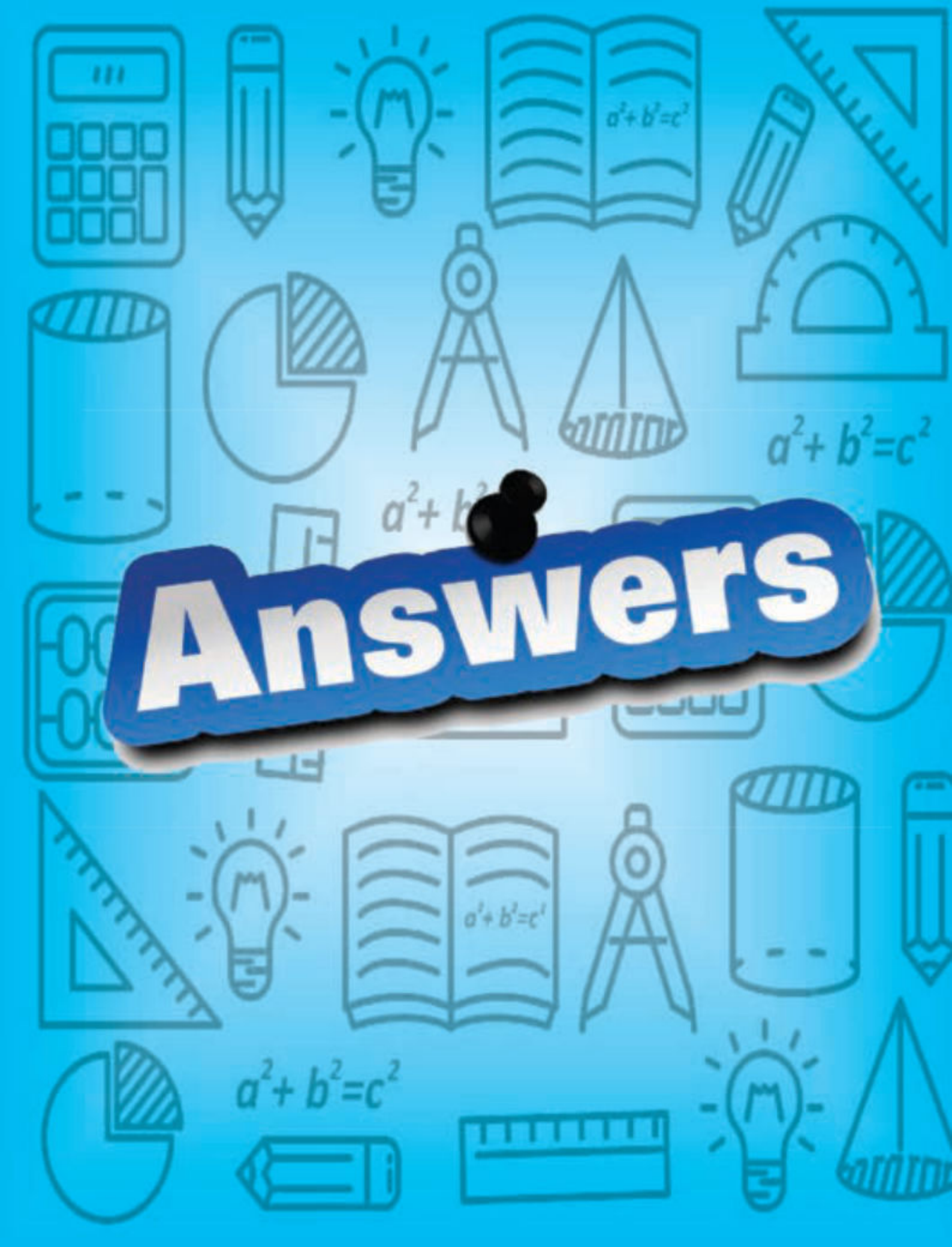


## Summary

- ◆ The distance between two points  $P_1(x_1, y_1)$  and  $P_2(x_2, y_2)$  is given by  

$$d = |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \text{ units.}$$
- ◆ The coordinates of the mid-point of line segment AB, passing through the points  $A(x_1, y_1)$  and  $B(x_2, y_2)$  is given by  $M(x, y) = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- ◆ Collinear points form a straight line.
- ◆ For collinearity of three points A, B and C,  
 either  $|AC| = |AB| + |BC|$ ,  
 or  $|AB| = |AC| + |CB|$   
 or  $|BC| = |BA| + |AC|$  holds good
- ◆ Three non-collinear points A, B and C form a triangle, if the sum of the lengths of any two sides is greater than the length of the third side.
- ◆ If  $|AB| + |BC| < |AC|$ , then no triangle can be formed by the points A, B and C.

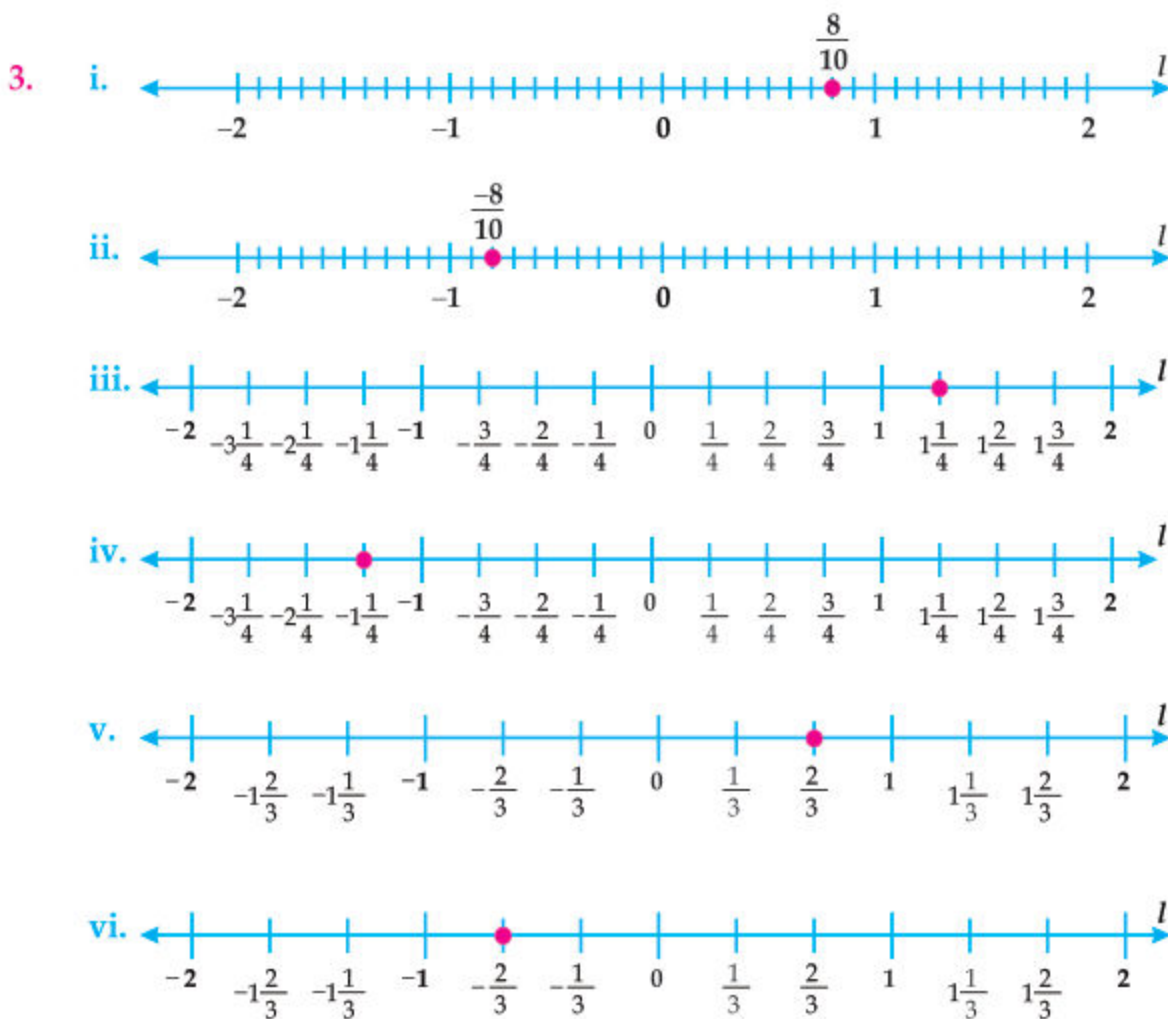


The background is a light blue gradient with a repeating pattern of white line-art icons. These icons include a calculator, a pencil, a lightbulb, an open book with the equation  $a^2 + b^2 = c^2$ , a set square, a cylinder, a pie chart, a compass, a cone, a protractor, and a ruler. The word "Answers" is written in a large, bold, white font with a blue drop shadow, tilted slightly upwards to the right. A black pushpin is pinned to the top center of the word.

# Answers

## Exercise 1.1

1. i. Rational number      ii. Irrational number  
 iii. Irrational number    iv. Rational number  
 v. Irrational number      vi. Irrational number  
 vii. Rational number      viii. Irrational number  
 ix. Irrational number     x. Rational number  
 xi. Irrational number    xii. Rational number
2. i. Terminating          ii. Non-terminating  
 iii. Non-terminating    iv. Terminating  
 v. Terminating          vi. Non-terminating



4. We can not make list of all real numbers between 1 and 2.
5. Pi ( $\pi$ ) is an irrational number because it is non terminating and non recurring decimal.
6. i. False ii. True iii. True iv. True v. True vi. False

### Exercise 1.2

1. i. Commutative property of addition.  
 ii. Associative property of addition.  
 iii. Left distributive property of multiplication over addition.  
 iv. Right distributive property of multiplication over addition.  
 v. Right distributive property of multiplication over subtraction.  
 vi. Commutative property of multiplication.  
 vii. Associative property of multiplication.  
 viii. Multiplicative inverse.  
 ix. Additive inverse.  
 x. Multiplicative inverse.  
 xi. Left distributive property of multiplication over subtraction.  
 xii. Multiplicative inverse.

2. i.  $\frac{\sqrt{2}}{5} + \frac{3}{\sqrt{6}} = \frac{\boxed{3}}{\sqrt{6}} + \frac{\sqrt{2}}{\boxed{5}}$       ii.  $\frac{7}{10} + \left( \frac{70}{\boxed{10}} + \frac{16}{33} \right) = \left( \frac{7}{\boxed{10}} + \frac{\boxed{70}}{10} \right) + \frac{16}{\boxed{33}}$   
 iii.  $\frac{99}{50} \times \frac{50}{99} = \boxed{1}$       iv.  $\frac{59}{95} \times \frac{95}{59} = \boxed{1}$   
 v.  $(-21) + (\boxed{21}) = 0$       vi.  $\frac{5}{8} \times \left( \frac{2}{3} + \frac{5}{7} \right) = \left( \frac{\boxed{5}}{8} \times \frac{2}{3} \right) + \left( \frac{5}{8} \times \frac{\boxed{5}}{7} \right)$
3. i.  $5 < 10$       ii.  $10 > 5$       iii.  $6 + 9$   
 iv.  $6 + 8$       v.  $6 + 6$
4. i.  $7 \times 12$       ii.  $5 \times 12$       iii.  $<$       iv.  $>$



5. Additive inverse

multiplicative inverse.

i.  $-3$

$\frac{1}{3}$

ii.  $7$

$-\frac{1}{7}$

iii.  $-0.3$

$\frac{1}{0.3}$

iv.  $\frac{\sqrt{5}}{5}$

$-\frac{5}{\sqrt{5}}$

v.  $\frac{-9}{\sqrt{12}}$

$\frac{\sqrt{12}}{9}$

vi.  $0$

does not exist

Exercise 1.3

1. i. Radicand = 5, Index = 3

ii. Radicand =  $\frac{x}{y}$ , Index = 4

iii. Radicand =  $x^2yz$ , Index = 5

iv. Radicand =  $ab$ , index = 2

v. Radicand =  $\frac{pq}{r}$ , index =  $n$



2. i.  $\left(\frac{3}{4}\right)^{\frac{1}{2}}$  ii.  $\left(\frac{x}{y}\right)^{\frac{5}{2}}$  iii.  $\left(\frac{x}{y}\right)^{\frac{5}{3}}$  iv.  $(yz)^{\frac{7}{3}}$  v.  $(27)^{\frac{1}{9}}$   
vi.  $(-64)^{\frac{2}{3}}$  vii.  $\left(\frac{1}{2}\right)^{\frac{m}{3}}$  viii.  $(xy)^{\frac{3}{5}}$  ix.  $\left(\frac{4}{3}\right)^{\frac{1}{6}}$

3. i.  $\sqrt[7]{(5)^3}$  ii.  $\sqrt[3]{\frac{a}{b^2}}$  iii.  $\sqrt[7]{\left(\frac{5}{7}\right)^{15}}$  iv.  $\sqrt{\left(\frac{a}{b}\right)^m}$  v.  $\sqrt[5]{\left(\frac{11}{13}\right)\left(\frac{12}{13}\right)}$

## Exercise 1.4

1. i. 27 ii. 20 iii.  $(a+b)(c+d)$   
2. i.  $\left(\frac{1}{3}\right)^9$  ii.  $\left(\frac{3}{4}\right)^7$  iii.  $\left(\frac{4}{5}\right)^8$  iv.  $-3^3 \times 5^6$  v.  $3^3 \times 4^6$   
vi.  $\frac{a^9}{b^9 c^9}$  vii.  $\frac{c^{10}}{d^{10}}$  viii.  $m^6 n^5 t^{11}$  ix.  $a^9 b^6 c^8$

3. i.  $5^6$  ii.  $x^{15} y^{15}$  iii.  $(4)^{10}$  iv.  $-3^9 \times 4^6$  v.  $\frac{b^6}{5^3}$   
vi.  $\frac{(4)^6}{9^6}$  vii.  $z^{24}$  viii.  $m^{100}$  ix.  $-(0.1)^{18}$

## Exercise 1.5

1. i.  $1+2i$  ii.  $2+2i$  iii.  $4i$   
iv.  $-1+i$  v.  $-2$  vi.  $-3+4i$

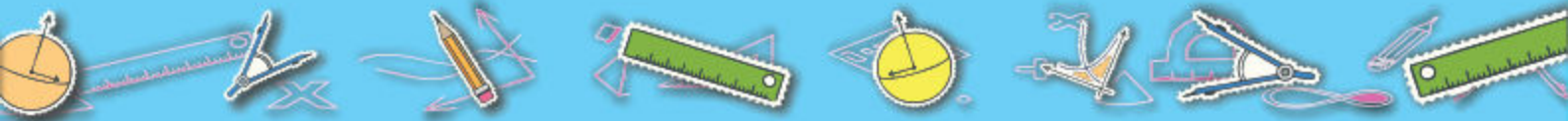


2. i.  $\operatorname{Re}(z)=1, \operatorname{Im}(z)=2$     ii.  $\operatorname{Re}(z)=4, \operatorname{Im}(z)=9$   
 iii.  $\operatorname{Re}(z)=-5, \operatorname{Im}(z)=6$     iv.  $\operatorname{Re}(z)=-1, \operatorname{Im}(z)=-1$   
 v.  $\operatorname{Re}(z)=\frac{-3}{4}, \operatorname{Im}(z)=\frac{4}{5}$     vi.  $\operatorname{Re}(z)=-1, \operatorname{Im}(z)=2$
3. i.  $\bar{z}=3-2i$     ii.  $\bar{z}=(0, 7)$     iii.  $\bar{z}=(-1, 0)$   
 iv.  $\bar{z}=1+i$     v.  $\bar{z}=\frac{-3}{4}+\frac{4}{5}i$     vi.  $\bar{z}=1-3i$
5. i.  $x=-5, y=5$     ii.  $x=\pm \frac{4}{3}$   
 $y=\pm \frac{3}{5}$   
 iii.  $x=\frac{-27}{5}, y=\pm 11$     iv.  $x=\frac{9\sqrt{30}}{\sqrt{5}}, y=\frac{-4}{27}$

### Exercise 1.6



1. i.  $(12, 5)$     ii.  $(\frac{13}{6}, \frac{13}{6})$     iii.  $(5, 21)$   
 iv.  $(0, -\frac{1}{15})$     v.  $(5, 0)$     vi.  $(0, -41)$   
 vii.  $(\frac{3-6\sqrt{2}}{4}, \frac{3+3\sqrt{2}}{2\sqrt{2}})$     viii.  $(\frac{-5}{13}, \frac{-27}{13})$
2. i.  $\frac{-1}{2}+\frac{1}{2}i$     ii.  $-4$     iii.  $-\frac{i}{2}$     iv.  $16$



## Review Exercise 1

1. i.  $\frac{1}{\sqrt{5}}$       ii. Set of real numbers      iii. 0  
 iv. 7      v. 0      vi. Irrational  
 vii. rational      viii.  $-3-5i$       ix. 2  
 x.  $(ac - bd, ad + bc)$

2. i. True      ii. True      iii. True      iv. True      v. True

3. i.  $a$       ii.  $b$       iii.  $c$       iv.  $b$

4. i. 2      ii.  $\frac{1}{3}$       5. i.  $3^{21}$       ii.  $2^{36}$

6. i. 7      ii. -1      iii.  $7+i$       iv.  $5\sqrt{2}$       v.  $\frac{7}{50} + \frac{1}{50}i$   
 vi.  $\frac{1}{5\sqrt{2}}$       vii.  $1+7i$       viii.  $-1+7i$

## Exercise 2.1

1. i.  $9.7 \times 10^3$       ii.  $4.98 \times 10^6$       iii.  $9.6 \times 10^7$       iv.  $4.169 \times 10^3$   
 v.  $8.4 \times 10^4$       vi.  $7.18 \times 10^{-1}$       vii.  $6.43 \times 10^{-3}$       viii.  $7.4 \times 10^{-3}$   
 ix.  $2.1005 \times 10^{-1}$
2. i. 70000      ii. 0.0000000008072      iii. 6018000      iv. 786500000  
 v. 0.000205      vi. 72500000000      vii. 4502000      viii. 0.00000002865  
 ix. 3056000

## Exercise 2.2

1. i.  $\log_7 343 = 3$       ii.  $\log_3 \frac{1}{81} = -4$       iii.  $\log_{10}(0.001) = -3$   
 iv.  $\log_8(4) = \frac{2}{3}$
2. i.  $(27)^{\frac{4}{3}} = 81$       ii.  $(2)^{-3} = \frac{1}{8}$       iii.  $10^0 = 1$       iv.  $(10)^{-2} = 0.01$
3. i.  $x = 4\sqrt{2}$       ii.  $a = 9$       iii.  $y = 4$       iv.  $x = 8$   
 v.  $y = 2$       vi.  $a = 4$       vii.  $a$  is any positive real number  
 viii.  $y = 1$       ix.  $x = 1$

## Exercise 2.3

- |                                               |                                             |                                              |
|-----------------------------------------------|---------------------------------------------|----------------------------------------------|
| 1. i. Characteristic : 0<br>Mantissa : 0.9031 | ii. Characteristic : 3<br>Mantissa : 0.7036 | iii. Characteristic : 0<br>Mantissa : 0.9997 |
| iv. Characteristic : 2<br>Mantissa : 0.8839   | v. Characteristic : -3<br>Mantissa : 0.5172 | vi. Characteristic : -5<br>Mantissa : 0.4771 |
| 2. i. 0.9542                                  | ii. 1.7448                                  | iii. 1.4711                                  |
| v. $\overline{3.6712}$                        | vi. $\overline{5.8808}$                     | iv. 2.6078                                   |
| 3. i. 0.4926                                  | ii. 2.4926                                  | iii. $\overline{3.4926}$                     |
| v. 2.4926                                     | vi. 5.4926                                  | iv. 3.4926                                   |

## Exercise 2.4

- |                   |                 |                  |
|-------------------|-----------------|------------------|
| 1. i. 3692        | ii. 0.5530      | iii. 2.278       |
| iv. 653800        | v. 0.0002425    | vi. 8.292        |
| 2. i. 2.954242509 | ii. 1.658393026 | iii. 4.563267445 |
| iv. 2.917137753   | v. -2.07007044  | vi. -4.013228266 |
| 3. i. 56.2989     | ii. 4.5803      | iii. 0.024367    |
| iv. 3019.95       | v. 0.0000000991 | vi. 1.8471       |

## Exercise 2.5

- |                                                                        |                                                 |
|------------------------------------------------------------------------|-------------------------------------------------|
| 1. i. $\log_a x + \log_a y + \log_a z$                                 | ii. $2\log_a x + \log_a y$                      |
| iii. $\log_a x + \log_a y - \log_a z$                                  | iv. $\frac{1}{2}\log_a x + \frac{1}{2}\log_a y$ |
| v. $-\frac{1}{2}\log_a x - \frac{1}{2}\log_a y - \frac{1}{2}\log_a z$  | vi. $3\log_a x + \log_a y - 2\log_a z$          |
| vii. $\frac{1}{2}\log_a x + \log_a y + \frac{1}{2}\log_a z$            | viii. $\frac{-7}{12}\log_a(x) - \log_a y$       |
| ix. $-\frac{2}{3}\log_a x + \frac{3}{2}\log_a y - \frac{2}{3}\log_a z$ |                                                 |
| 2. i. $\log_a(2\sqrt{2})$                                              | ii. $\log_a(x^2 - 1)$                           |
|                                                                        | iii. $\log \frac{(x+1)^2}{x(x-1)}$              |

3. i. 1.1761      ii. 1.8062      iii. 0.5      iv. 1.6812  
      v. 0.6276      vi. 1.4771      vii. 0.4260      viii. 0.4604

### Exercise 2.6

1. i. 253.688      ii. 6750      iii. 48.2176      iv.  $x = 930.80$   
      v. 15.20      vi. 1.2585      vii. 410130      viii.  $1.84077 \times 10^{13}$
2. i. 8      ii. 22      iii. 15      iv. 14      v. 29

### Review Exercise 2

1. i. False      ii. False      iii. True      iv. False      v. True
2. i. Common logarithm      ii. 0      iii. Mantissa  
      iv. 9      v.  $\log_a n$       vi.  $a^x = y$   
      vii.  $\log_a y = 10$       viii. 1      ix.  $\log_a m - \log_a n$   
      x. 2      xi. 0      xii. -5
3. i.  $d$       ii.  $b$       iii.  $b$       iv.  $b$       v.  $a$   
      vi.  $c$       vii.  $c$       viii.  $b$       ix.  $b$       x.  $c$

### Exercise 3.1

1. i. Polynomial      ii. Not a polynomial      iii. Polynomial  
      iv. Not a polynomial      v. Not a polynomial      vi. Not a polynomial
2. i. Rational      ii. Not a rational      iii. Rational  
      iv. Not a rational      v. Rational      vi. Not a rational
3. i.  $p-10$       ii.  $\frac{a}{a+b}$       iii.  $\frac{a}{2(a+b)}$   
      iv.  $x+y-z$       v.  $\frac{3m(m+5)}{2}$       vi.  $\frac{x-3}{x-2}$



4. i.  $\frac{4x^2-1}{x^2-1}$  ii.  $\frac{3x+7}{(x+2)(x+3)}$  iii.  $\frac{2x^2y^2+xy+1}{(xy+1)(xy-1)}$   
 iv.  $\frac{-15}{(x+3)(x+6)}$  v.  $\frac{-2b}{a^2-b^2}$  vi.  $\frac{-(y-1)}{y+1}$
5. i.  $\frac{8y^3}{(2y-x)^2(2y+x)}$  ii.  $-\frac{2x+3y}{y}$  iii. 1  
 iv.  $\frac{5}{3}$  v.  $\frac{(q-5)(q+3)}{q^2}$  vi.  $\frac{8(z-1)}{z-5}$
6.  $\frac{2(x^2+y^2)}{x^2-y^2}$
7. i.  $\frac{1}{6}$  ii.  $9\frac{9}{55}$  iii.  $-\frac{17}{73}$   
 iv.  $-4\frac{4}{9}$  v.  $1\frac{1}{11}$

### Exercise 3.2



1.  $a^2+b^2=50, ab=7$  2.  $a^2+b^2=17, ab=4$  3.  $a^2+b^2+c^2=55$   
 4.  $a^2+b^2+c^2=\frac{5}{9}$  5.  $a+b+c=\pm 7$  6.  $a+b+c=\pm\sqrt{2.5}$   
 7.  $ab+bc+ca=40$  8.  $a^3+b^3=28$  9.  $ab=-8$   
 10.  $ab=-4$  11.  $a^3-b^3=230$  12.  $125x^3+y^3=247$   
 13.  $216a^3-343b^3=12419$  14.  $x^3+\frac{1}{x^3}=322$  15.  $x^3-\frac{1}{x^3}=1364$





16. i.  $\frac{27b^3}{8} + \frac{8}{27b^3}$

ii.  $\frac{343y^6}{729} + \frac{729}{343y^6}$

iii.  $\frac{x^{12}}{1728} - \frac{1728}{x^{12}}$

iv.  $c^6 - \frac{1}{c^6}$

17. i.  $8x^6 + 27y^6$

ii.  $8x^6 - 27y^6$

iii.  $x^{12} - y^{12}$

iv.  $256x^8 - 6561y^8$

### Exercise 3.3

1. i.  $\frac{3z}{x^2}$  ii.  $4\sqrt[3]{4a^2b^4c^3}$  iii. 2 iv.  $36\sqrt{6}$

v.  $\frac{25}{32}$  vi.  $\frac{14\sqrt{3}}{11}$  vii. 6 viii. 2

2. i.  $8 + 4\sqrt{3}$  ii.  $6\sqrt{6} - 2\sqrt{3}$  iii.  $8\sqrt{12} - \sqrt{8}$  iv.  $2 + \sqrt{3}$

3. i.  $66\sqrt{2}$  ii.  $13\sqrt{5}$  iii.  $20 + 9\sqrt{3}$  iv.  $15\sqrt{10}$

v.  $4\sqrt{5} + 25$  vi.  $\sqrt{11}$  vii. 136 viii.  $3\sqrt{2}$

ix.  $\frac{1}{3}$  x. 2 xi.  $134 - 24\sqrt{30}$  xii.  $30 + 12\sqrt{6}$

### Exercise 3.4

1. i.  $2 - \sqrt{3}$  ii.  $3 - 2\sqrt{2}$  iii.  $-\left(\frac{5\sqrt{2} + 4\sqrt{3}}{2}\right)$

iv.  $16(2\sqrt{3} - 11)$  v.  $\frac{83 - 18\sqrt{2}}{79}$  vi.  $\frac{11 + 3\sqrt{3}}{2}$



2. i.  $\left(x + \frac{1}{x}\right)^2 = 256$  ii.  $x = -\left(\frac{4\sqrt{7} + 11}{9}\right)$

iii.  $x + \frac{1}{x} = 6, x - \frac{1}{x} = -4\sqrt{2}, x^2 + \frac{1}{x^2} = 34, x^2 - \frac{1}{x^2} = -24\sqrt{2}, x^4 + \frac{1}{x^4} = 1154$

3. 322

4. 194

5.  $112\sqrt{3}$

### Review Exercise 3

1. i.  $b$

ii.  $b$

iii.  $a$

iv.  $a$

v.  $a$

vi.  $a$

vii.  $b$

viii.  $a$

ix.  $a$

x.  $a$

xi.  $a$

xii.  $b$

2. i. Highest power or exponent on the variable

ii.  $2 + \sqrt{3}$

iii. 4

iv. Irrational expression

v.  $x^4 - y^4$

### Exercise 4.1



1. i.  $4(x + 4y + 6z)$

ii.  $x^2(1 + 3y + 4y^2z)$

iii.  $3pq(r + 2t + s)$

iv.  $9qr(s^2 + t^2)(1 + 2qr)$

v.  $\frac{xz^2}{4} \left( \frac{1}{4} - \frac{x}{2} + \frac{xz}{3} \right)$

vi.  $a(x - y)(1 - ab + ab^2)$

2. i.  $(7 + z)(x + z)$

ii.  $3(3ab - 2c)(a + 2b)$

iii.  $2(t - 2p)(3 + 2q)$

iv.  $(r + 9s)(r - 7s)$







v.  $(1-z)\left(\frac{y^2}{4} - \frac{z^2t}{9}\right)$

3. i.  $(2a+3b)^2$

iii.  $\left(x + \frac{1}{2x}\right)^2$

v.  $(25+a^2b)^2$

4. i.  $(b^2-2c^2)^2$

iii.  $2ab^3(a-4b)^2$

v.  $(xy-0.05)^2$

5. i.  $(2a-3b)(2a+3b)$

iii.  $(10xz+y^2)(10xz-y^2)$

v.  $\left(\frac{8f}{9} - \frac{9g^2}{8}\right)\left(\frac{8f}{9} + \frac{9g^2}{8}\right)$

6. i.  $8xz$

ii.  $4(3a-2b)(a-7b)$

iii.  $(13x^2-3t-4)(13x^2+3t+4)$

iv.  $(13x^2-5y^2)(5x^2-3y^2)$

v.  $\left(a + \frac{1}{a} + b - \frac{1}{b}\right)\left(a + \frac{1}{a} - b + \frac{1}{b}\right)$

vi.  $\left(3x + \frac{1}{3x} + 2y - \frac{1}{2y}\right)\left(3x + \frac{1}{3x} - 2y + \frac{1}{2y}\right)$

7. i.  $(x+y-3z^2)(x+y+3z^2)$

iii.  $(4d^2-c^2+d)(4d^2+c^2-d)$

vi.  $\frac{1}{11}(2y+z)(5x-7y)$

ii.  $(6x^2+1)^2$

iv.  $(9y+8z)^2$

vi.  $(a+0.2)^2$

ii.  $\left(\frac{3x^2}{2} - \frac{2}{3x^2}\right)^2$

iv.  $(3p+3q-r^2)^2$

vi.  $(a-b-9)^2$

ii.  $(4x-5y)(4x+5y)$

iv.  $\left(\frac{x^2}{10} + 10y^2\right)\left(\frac{x^2}{10} - 10y^2\right)$

vi.  $\left(\frac{x^2}{11} - 11y\right)\left(\frac{x^2}{11} + 11y\right)$

ii.  $(2a+2b^2-3c)(2a+2b^2+3c)$

iv.  $(2x+2y^2+3y^3)(2x+2y^2-3y^3)$



- v.  $(x+y-1)(x-y-z)$  vi.  $(2x+y+1)(2x-y-1)$
8. i.  $(\sqrt{ab}-\sqrt{c})(\sqrt{ab}+\sqrt{c})$  ii.  $(2\sqrt{x}-3\sqrt{y})(2\sqrt{x}+3\sqrt{y})$
- iii.  $\left(\sqrt{yz}-\frac{1}{\sqrt{yz}}\right)\left(\sqrt{yz}+\frac{1}{\sqrt{yz}}\right)$  iv.  $\left(\sqrt{xzt}-\frac{1}{\sqrt{t}}\right)\left(\sqrt{xzt}+\frac{1}{\sqrt{t}}\right)$

## Exercise 4.2



1. i.  $(a^2+x^2+ax)(a^2+x^2-ax)$  ii.  $(b^2-b+1)(b^2+b+1)$
- iii.  $(a^2+x^2-ax)(a^2+x^2+ax)(a^4+x^4-a^2x^2)$
- iv.  $(z^2+z+1)(z^2-z+1)(z^4-z^2+1)$
2. i.  $(x^2+2xy+2y^2)(x^2-2xy+2y^2)$  ii.  $9(2x^2z^2+2xyz+y^2)(2x^2z^2-2xyz+y^2)$
- iii.  $(2t^2+10t+25)(2t^2-10t+25)$  iv.  $(2t^2+2t+1)(2t^2-2t+1)$
3. i.  $(x-2)(x+5)$  ii.  $(ab+2)(ab-5)$
- iii.  $(y-7)(y+14)$  iv.  $(xyz-4)(xyz+6)$
4. i.  $(3y+8z)(3y-z)$  ii.  $2(7x+1)(3x-1)$
- iii.  $(2x+1)(2x+5)$  iv.  $(3x+y)(x-13y)$

## Exercise 4.3



1. i.  $(x-2)^2(x-7)(x+3)$  ii.  $(x^2+5x+3)(x^2+5x+7)$
- iii.  $(x-3)(x+1)(x^2-2x+10)$  iv.  $(x^2-8x+1)(x^2-8x-1)$
- v.  $(x^2+9x-2)(x^2+9x+6)$  vi.  $(x-6)(x+1)(x^2-5x+16)$



2. i.  $(x^2 + 5x - 2)(x^2 + 5x + 12)$  ii.  $(x^2 + 7x + 16)(x + 6)(x + 1)$   
 iii.  $(x^2 - 5x + 15)(x^2 - 5x - 5)$  iv.  $(x^2 - 12x + 30)(x - 8)(x - 4)$   
 v.  $(x^2 - 5x - 10)(x^2 - 5x + 20)$  vi.  $(x^2 - 7x + 27)(x^2 - 7x - 5)$
3. i.  $(x + \sqrt{3})(x - \sqrt{3})(x + 2\sqrt{3})(x - 2\sqrt{3})$   
 ii.  $(x^2 + 1)(x + \sqrt{14})(x - \sqrt{14})$  iii.  $(x + \sqrt{3})^2(x - \sqrt{3})^2$   
 iv.  $(x + \sqrt{2})(x - \sqrt{2})(x + 4\sqrt{2})(x - 4\sqrt{2})$   
 v.  $(x + \sqrt{5})(x - \sqrt{5})(x + 2\sqrt{5})(x - 2\sqrt{5})$  vi.  $(x + 2\sqrt{3})(x - 2\sqrt{3})(x^2 - 2x - 12)$

## Exercise 4.4

1. i.  $(b + c)^3$  ii.  $(2x + y)^3$  iii.  $\left(4x + \frac{1}{4}\right)^3$   
 iv.  $(2x + 3)^3$  v.  $\left(\frac{1}{3} + y^2\right)^3$  vi.  $\left(\frac{2}{3}x + \frac{3}{2}y\right)^3$   
 vii.  $\left(\frac{4}{3} + x\right)^3$  viii.  $\left(\frac{z}{2} + \frac{y}{3}\right)^3$
2. i.  $(d - 2c)^3$  ii.  $\left(x^2 - \frac{4}{3}\right)^3$  iii.  $\left(\frac{x}{5} - y\right)^3$   
 iv.  $(5z - y^2)^3$  v.  $\left(\frac{z}{3} - 6y\right)^3$  vi.  $\left(\frac{b^2}{3} - \frac{c^2}{2}\right)^3$   
 vii.  $\left(6 - \frac{z}{2}\right)^3$  viii.  $\left(\frac{2}{3}x - \frac{3}{2}y\right)^3$

## Exercise 4.5

1. i.  $(x + 2y)(x^2 - 2xy + 4y^2)$  ii.  $a^2(a + b)(a^2 - ab + b^2)(a^6 - a^3b^3 + b^6)$   
 iii.  $(a^2 + 1)(a^4 - a^2 + 1)$  iv.  $(ab + 8)(a^2b^2 - 8ab + 64)$

- v.  $b^3(a+3b)(a^2-3ab+9b^2)$  vi.  $\left(\frac{x}{5} + \frac{5}{x}\right)\left(\frac{x^2}{25} - 1 + \frac{25}{x^2}\right)$   
 vii.  $x^3(x^2+y^2z^3)(x^4-x^2y^2z^3+y^4z^6)$   
 viii.  $\left(\frac{x^2}{3} + \frac{2}{x}\right)\left(\frac{x^4}{9} - \frac{2x}{3} + \frac{4}{x^2}\right)$   
 2. i.  $(x-2y)(x^2+2xy+4y^2)$  ii.  $(x^3-2y^3)(x^6+2x^3y^3+4y^6)$   
 iii.  $\left(10 - \frac{xy}{5}\right)\left(100 + xy + \frac{x^2y^2}{25}\right)$   
 iv.  $(a+b)(a^2-ab+b^2)(a-b)(a^2+ab+b^2)$   
 v.  $\left(\frac{x}{2} + \frac{2}{x^2}\right)\left(\frac{x^2}{4} - \frac{1}{x} + \frac{4}{x^4}\right)\left(\frac{x}{2} - \frac{2}{x^2}\right)\left(\frac{x^2}{4} + \frac{1}{x} + \frac{4}{x^4}\right)$   
 vi.  $(x-y)(x+y)(x^2+y^2)(x^4+y^4-x^2y^2)(x^2+y^2+xy)(x^2+y^2-xy)$   
 vii.  $\left(\frac{3}{x} - 2y^2\right)\left(\frac{9}{x^2} + \frac{6y^2}{x} + y^4\right)$  viii.  $\left(2x^2 - \frac{1}{9}\right)\left(4x^4 + \frac{2x^2}{9} + \frac{1}{81}\right)$

## Exercise 4.6

1. i.  $R = -2$  ii.  $R = 2$  iii.  $R = 18$  iv.  $R = -42$  v.  $R = -11$   
 vi.  $R = \frac{3}{2}$  vii.  $R = -8$  viii.  $R = 3y^4$   
 2.  $m = -1$  3.  $k = -24$  4.  $r = -1, r = 3$

## Exercise 4.7

1. i.  $R = 0, q(x) = x^2 + 1$  ii.  $R = -2, q(x) = x^2 - 2x + 1$   
 iii.  $R = -60, q(x) = x^2 - 8x + 27$  iv.  $R = 4, q(x) = x^2 + 8x + 5$   
 v.  $R = 29, q(x) = x^3 - 3x^2 + 7x - 15$  vi.  $R = 1, q(x) = x^3 + 2x^2 + x + 2$   
 vii.  $R = -291, q(x) = x^4 - 3x^3 + 10x^2 - 32x + 96$   
 viii.  $R = 174, q(x) = x^4 + 2x^3 + 7x^2 + 18x + 60$   
 ix.  $R = 175, q(x) = 2x^3 + 2x^2 + 104x + 40$

x.  $R = -300$ ,  $q(x) = 6x^3 - 42x^2 + 90x - 114$

2.  $k = 24$

3.  $m = 4$

4.  $m = -24$

5.  $m = -1$

### Exercise 4.8

1. i.  $(x-1)(x^2+1)$

ii.  $(x+1)^2(x-1)$

iii.  $(x-1)(x-2)(x-3)$

iv.  $(x+2)(x-2)(x+5)$

v.  $(x-2)(x^2+9)$

vi.  $(x+1)(2x-1)(3x+2)$

vii.  $(x+1)(x+3)(x+4)$

viii.  $(x+1)(2x+1)(x+3)$

ix.  $(x+2)(x+4)(x+6)$

### Review Exercise 4

1. i. True

ii. True

iii. True

iv. False

v. False

vi. False

2. i.  $4x + y^2$

ii.  $x^2 + 4xy + 16y^2$

iii.  $x + 3$

iv.  $2xy$

v.  $a^2 - 3ab + 9b^2$

3. i.  $b$

ii.  $c$

iii.  $d$

iv.  $a$

v.  $b$

vi.  $c$

### Exercise 5.1

1. i.  $\text{HCF} = 24x^3y^5z^2$

ii.  $\text{HCF} = 6r^3s^3t^3$

iii.  $\text{HCF} = (x+3)$

iv.  $\text{HCF} = (2x-3)$

v.  $\text{HCF} = 2(a-2b)$

vi.  $\text{HCF} = (x+1)$

2. i.  $\text{HCF} = (x+1)$

ii.  $\text{HCF} = (x^2 + 7x + 12)$

iii.  $\text{HCF} = (x-2)$

iv.  $x^2 + 3x + 1$

3. i.  $\text{LCM} = 81a^4b^5c^8$

ii.  $600p^5q^4r^8$

iii.  $7x(x-1)(3x-2)$

iv.  $(x+4)(x+7)(x-3)$

v.  $(3x+1)(x-1)(2x+3)$



- vi.  $(x+y)(x-y)(x^2+xy+y^2)(x^2-xy+y^2)$
4. i.  $(x-20)(x-5)(x+4)$  ii.  $(3x+2)(x+4)(2x-1)$   
 iii.  $(x+y+z)(x-y-z)(y-z-x)$  iv.  $12x^3(x-4)(x+7)(x-2)$
5.  $(x-8)(x^2-6x+6)$  6.  $x^2+2x-3$
7.  $9x^4+15x^3-12x-14x^2+8$  9. 6 cm
10. 11 : 54 am

## Exercise 5.2

1. i.  $\frac{7x+3}{(x+1)^2}$  ii.  $\frac{12x^2+29x+16}{3x(2x+1)(x+1)}$   
 iii.  $\frac{-4x^3-x^2-9x+4}{(x+1)^2(x-3)}$  iv.  $\frac{2(x^2+3x+3)}{(x+1)(x+2)(x+3)}$   
 v.  $2x+6$  vi.  $\frac{x+3}{x+9}$   
 vii.  $\frac{-(x+3)(4x+7)}{(x+1)^2(x+2)}$  viii.  $\frac{-2x+7}{(x-2)(x-3)}$   
 ix.  $\frac{x^2-5x-42}{(x^2-9)(x+4)(x+5)}$  x.  $\frac{2(x^2+5)}{4x^2+x+2}$

## Exercise 5.3

1. i.  $6x-5y$  ii.  $3x+\frac{1}{x}$  iii.  $2x^2y^2-\frac{3xy}{z^2}$   
 iv.  $18-12x-4y$  v.  $(x+\frac{1}{x}+1)$  vi.  $3(2x-1)(x-3)$   
 vii.  $(x-1)(x-3)$  viii.  $(x+3)(x+5)(x+2)$
2. i.  $x^2+x+1$  ii.  $5x^2+4x+1$  iii.  $2x^2+2x+4$




 iv.  $\left(\frac{x}{y} + 7 - \frac{y}{x}\right)$

v.  $x - 1 + \frac{1}{x}$

vi.  $x + \frac{y}{3} + 3z$

vii.  $x^2 - 4 + \frac{1}{x^2}$

viii.  $x^3 - 2 + \frac{1}{x^3}$

3. 7    4.  $-24x^2 + 9$     5.  $m = 20$     6.  $p = 56, q = 49$     7.  $a = 12, b = 9$

### Review Exercise 5

1. i. True    ii. False    iii. True    iv. False    v. True

2. i. two    ii.  $p(x)q(x)$     iii. 1

iv.  $(y+1)(y+2)(y+3)$     v.  $y + \frac{1}{y}$

3. i. d    ii. d    iii. b    iv. c    v. b    vi. d  
 vii. c    viii. b    ix. b    x. c

### Exercise 6.1

1. i.  $x = 20$     ii.  $x = -12$     iii.  $x = 30$     iv.  $x = 40$

v.  $y = \frac{1}{15}$     vi.  $y = \frac{11}{20}$     vii.  $x = \frac{44}{17}$     viii.  $x = 105$

ix.  $\frac{105}{13}$     x.  $x = 1$     xi.  $x = -\frac{20}{7}$     xii.  $x = 12$

xiii.  $x = -\frac{5}{4}$     xiv.  $x = -4$     xv.  $m = \frac{-11}{6}$

2. {5}    3.  $x = 7$     4. Bilal = 18 years old  
 Ali = 12 years old

5. i. {1}    ii. {10}    iii. {100}  
 iv. {-12}    v. {143}    vi. { }  
 vii. {2}    viii. {81}    ix. {80}

## Exercise 6.2

1. i.  $\left\{-\frac{7}{2}, \frac{5}{2}\right\}$  ii.  $\{1\}$  iii.  $\{-42, 42\}$  iv.  $\left\{-\frac{25}{2}, \frac{23}{2}\right\}$   
 v.  $\{3\}$  vi.  $\left\{-\frac{78}{5}, \frac{76}{5}\right\}$  vii.  $\left\{-\frac{23}{2}, \frac{17}{2}\right\}$  viii.  $\{-10, 6\}$   
 ix.  $\left\{-\frac{19}{14}, -\frac{13}{14}\right\}$  x.  $\{-41, 44\}$  xi.  $\{\}$  xii.  $\{-4, 3\}$

## Exercise 6.3

1. i.  $\{14, 15, 16, \dots\}$



- ii.  $\{x \mid x \in \mathbb{R} \wedge x > 4\}$



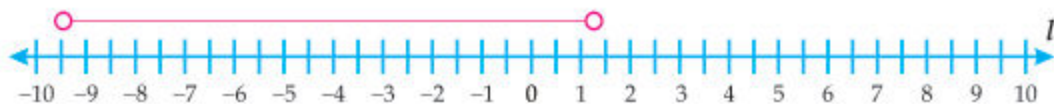
- iii.  $\{1, 2, 3, 4\}$



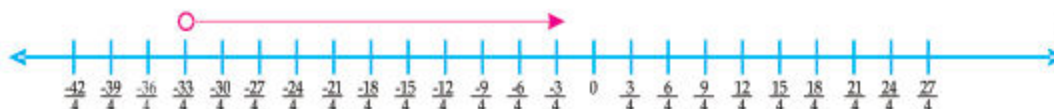
- iv.  $\{\}$



- v.  $\{y \mid y \in \mathbb{R} \wedge -\frac{19}{2} < y < \frac{3}{2}\}$



- vi.  $\{y \mid y \in \mathbb{R} \wedge y > -\frac{33}{4}\}$



2. All the numbers  $\geq 4$
3. Ali must score at least 87 to qualify for bonus prize.

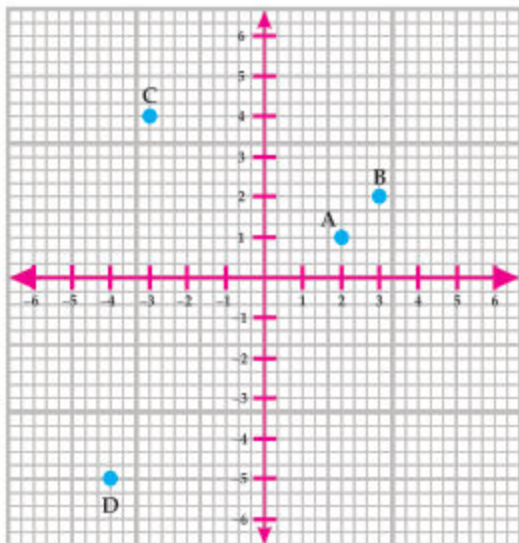
### Review Exercise 6

1. i. False    ii. False    iii. True    iv. True    v. False
2. i.  $\{0\}$     ii.  $\{20\}$     iii.  $\{\pm 4\}$
- iv.  $\{-1\}$     v.  $\{y \mid y \in \mathbb{R} \wedge -2 < y < 3\}$
3. i. a    ii. a    iii. c    iv. a    v. b    vi. c
- vii. c    viii. c    ix. c    x. a

### Exercise 7.1

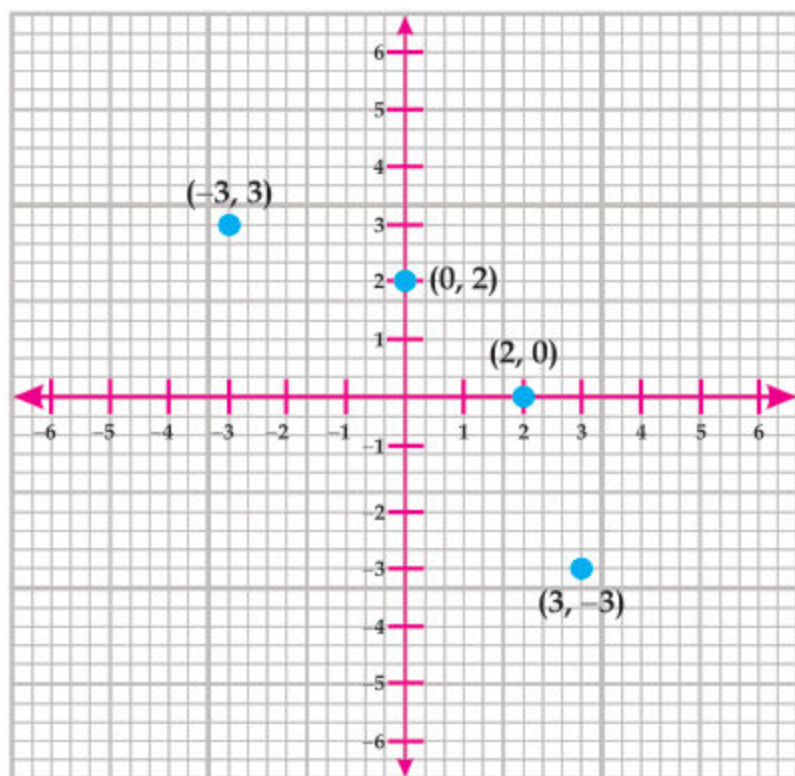
1. i. Abscissa = -2, ordinate = 2    ii. Abscissa = 5, ordinate = -1
- iii. Abscissa = 4, ordinate = 0    iv. Abscissa = -5, ordinate = -6
- v. Abscissa = 3, ordinate = 4    vi. Abscissa =  $-\sqrt{8}$ , ordinate =  $\sqrt{8}$
2. i. Lies in quadrant \_ IV    ii. Lies in quadrant \_ II
- iii. Lies in quadrant - IV    iv. Lies in quadrant - III
- v. Lies in quadrant - I    vi. Lies in quadrant - I

3. i.

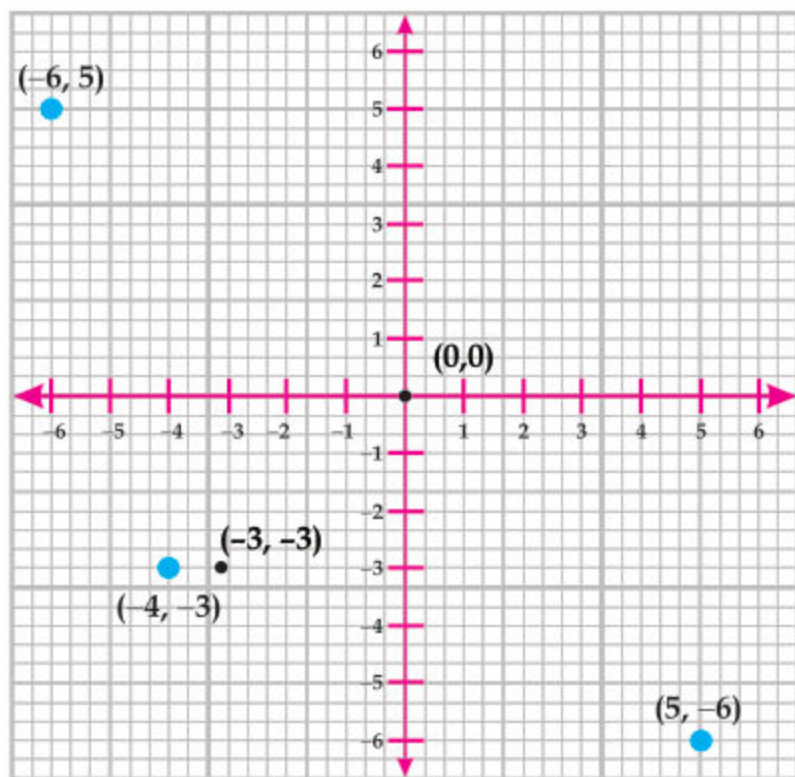




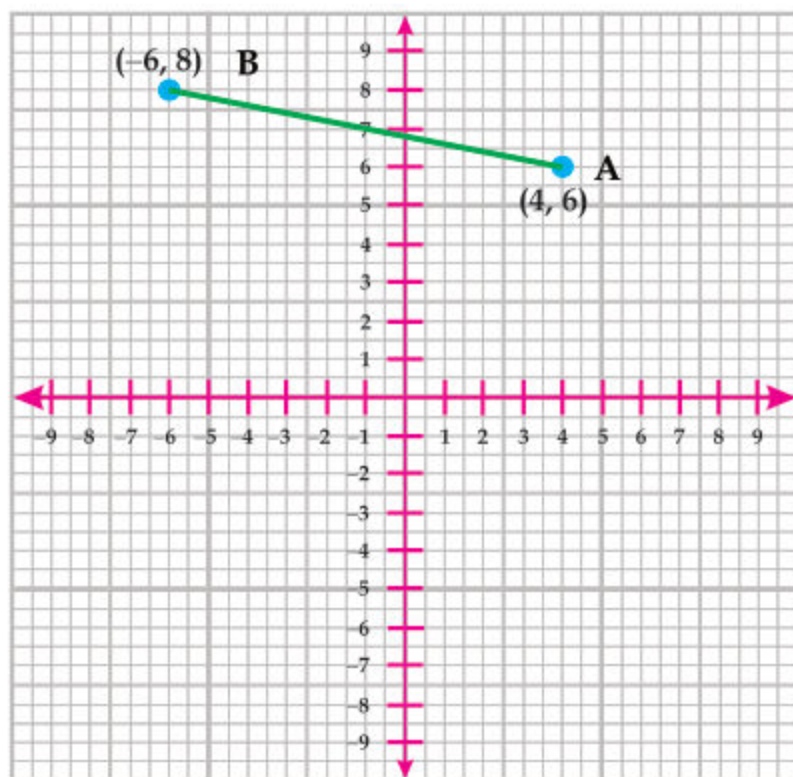
ii.



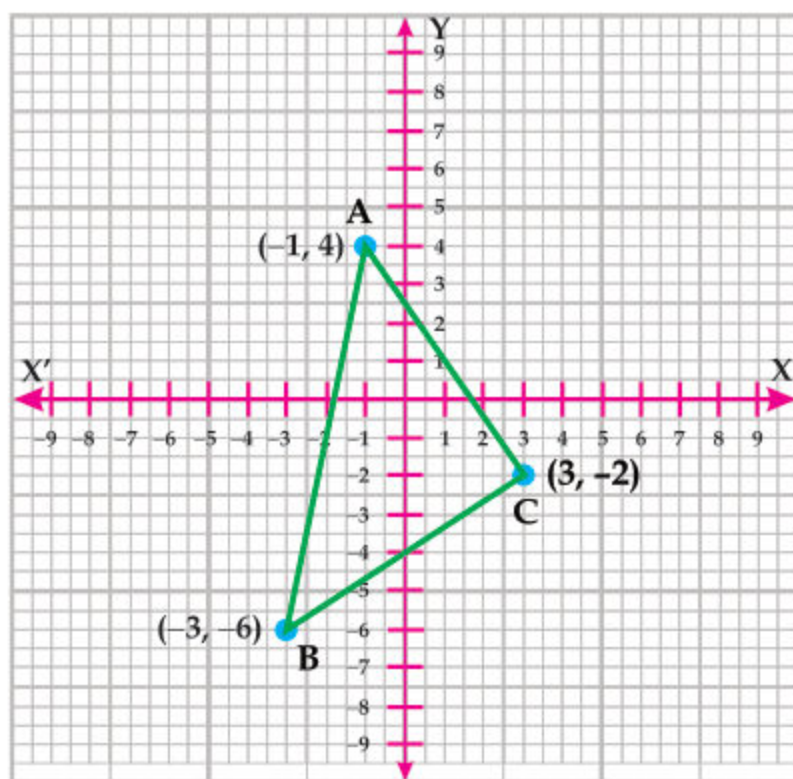
iii.



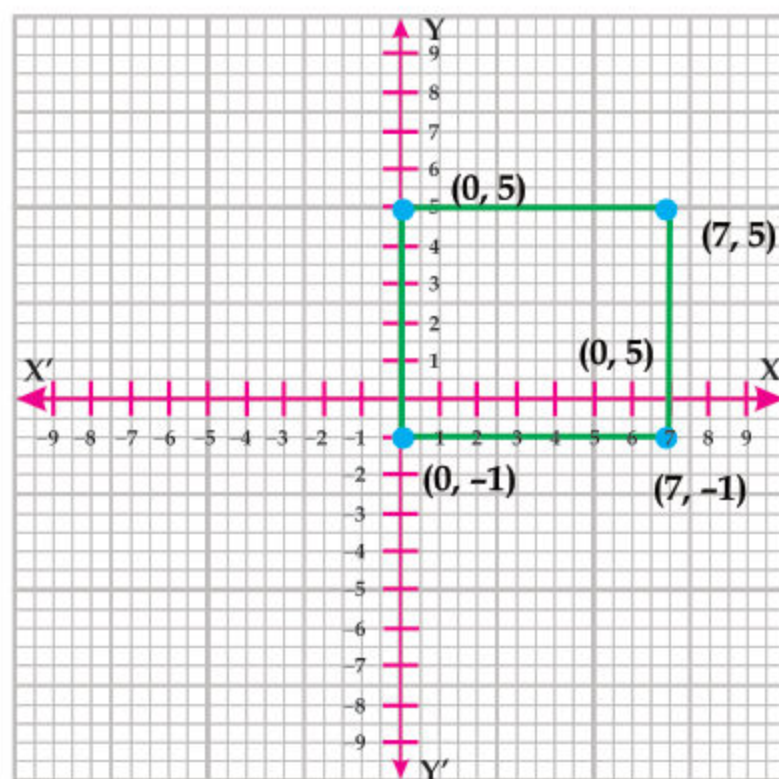
4.



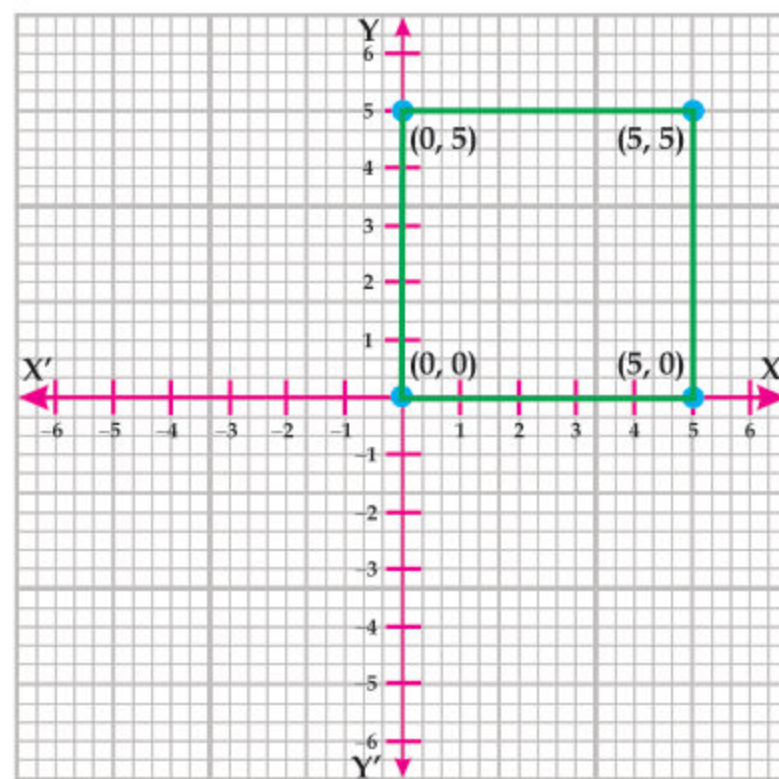
5.



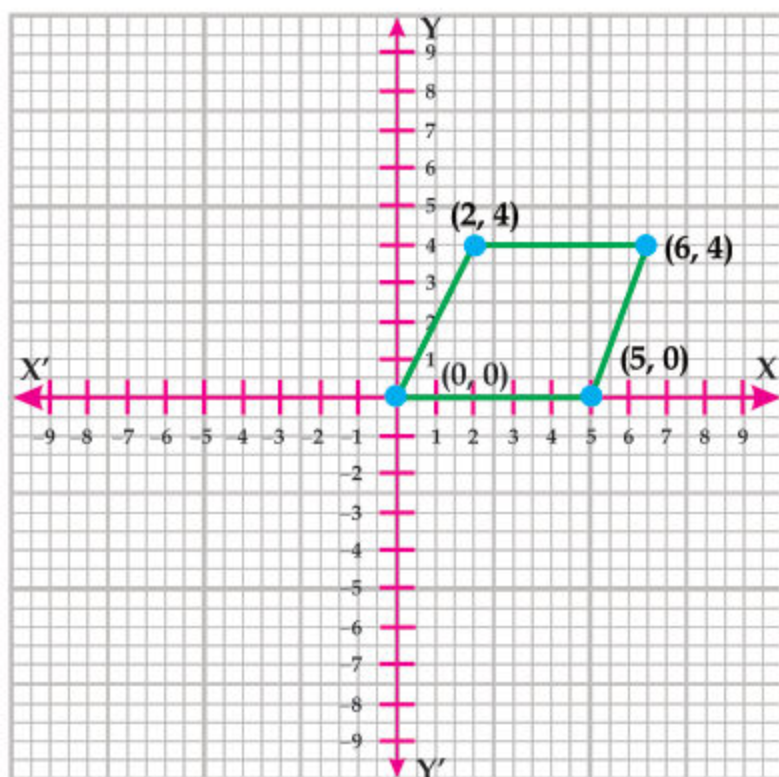
6.



7.



8.

9. i.  $y = 2 - x$ 

|     |    |    |    |   |   |   |    |
|-----|----|----|----|---|---|---|----|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3  |
| $y$ | 5  | 4  | 3  | 2 | 1 | 0 | -1 |

ii.  $y = 2x - 6$ 

|     |     |     |    |    |    |    |   |
|-----|-----|-----|----|----|----|----|---|
| $x$ | -3  | -2  | -1 | 0  | 1  | 2  | 3 |
| $y$ | -12 | -10 | -8 | -6 | -4 | -2 | 0 |

iii.  $x = 12 - 2y$ 

|     |    |    |    |    |    |   |   |
|-----|----|----|----|----|----|---|---|
| $y$ | -3 | -2 | -1 | 0  | 1  | 2 | 3 |
| $x$ | 18 | 16 | 14 | 12 | 10 | 8 | 6 |

iv.  $x = 2y - 3$ 

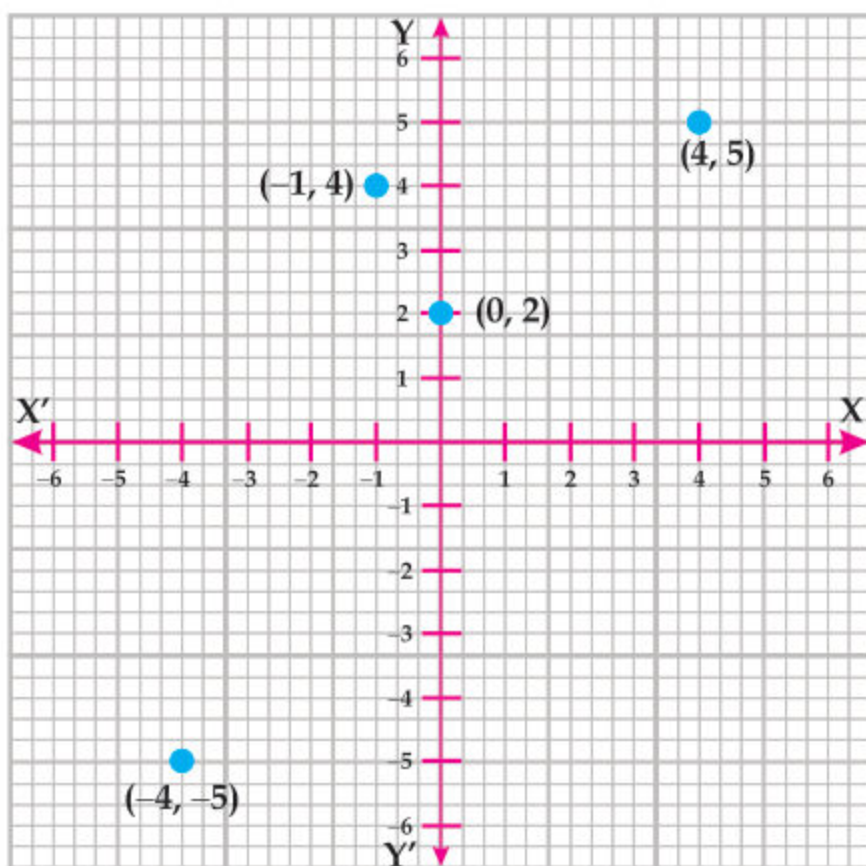
|     |    |    |    |    |    |   |   |
|-----|----|----|----|----|----|---|---|
| $y$ | -3 | -2 | -1 | 0  | 1  | 2 | 3 |
| $x$ | -9 | -7 | -5 | -3 | -1 | 1 | 3 |

## Exercise 7.2

1.

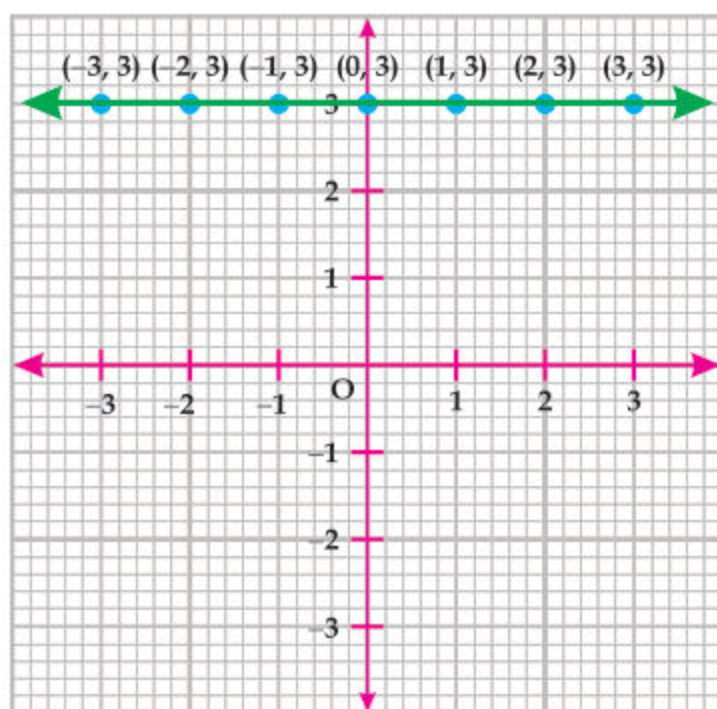
$$y = 6 - x$$

| $y$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|-----|----|----|----|---|---|---|---|
| $x$ | 9  | 8  | 7  | 6 | 5 | 4 | 3 |



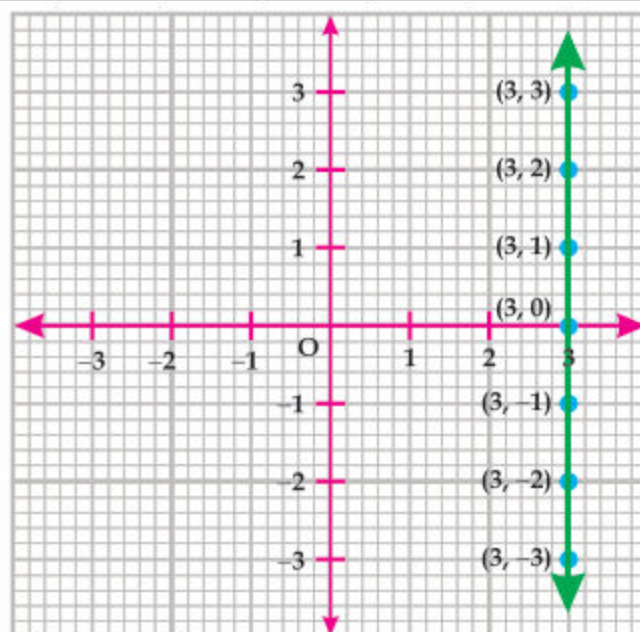
3. i.

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 3  | 3  | 3  | 3 | 3 | 3 | 3 |



ii.

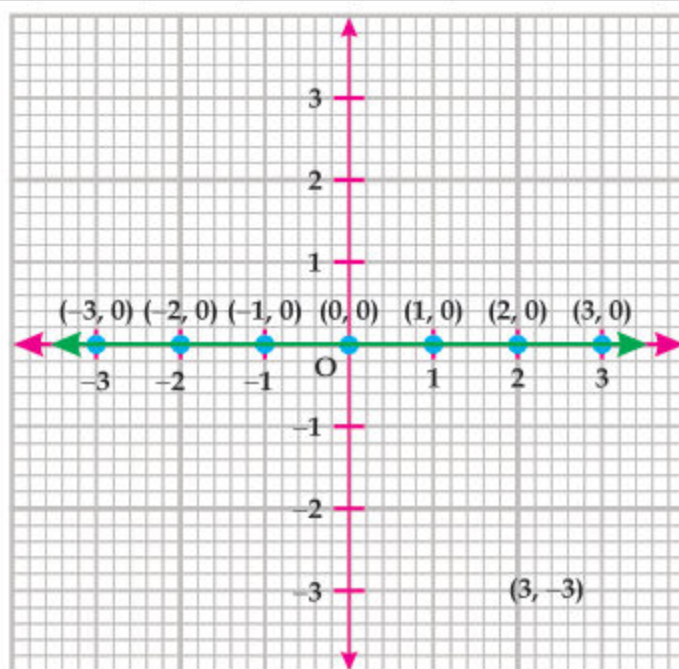
|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | 3  | 3  | 3  | 3 | 3 | 3 | 3 |
| $y$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |





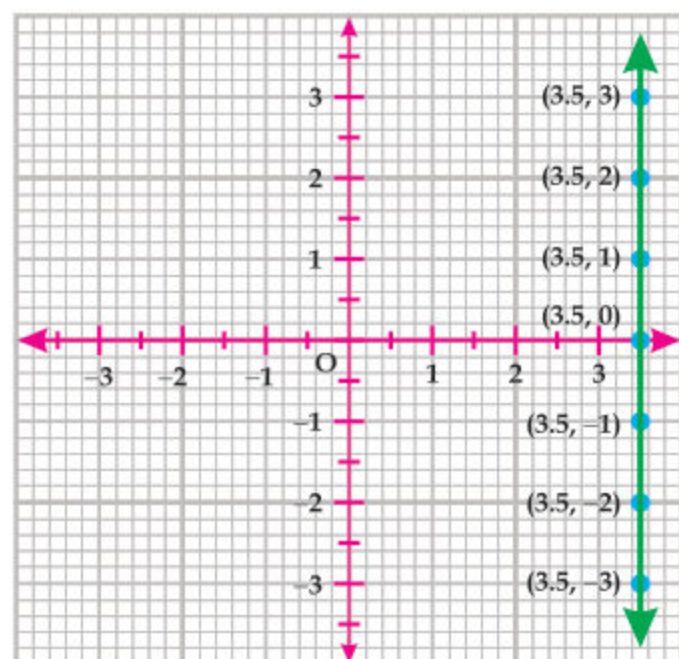
iii.

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 0  | 0  | 0  | 0 | 0 | 0 | 0 |



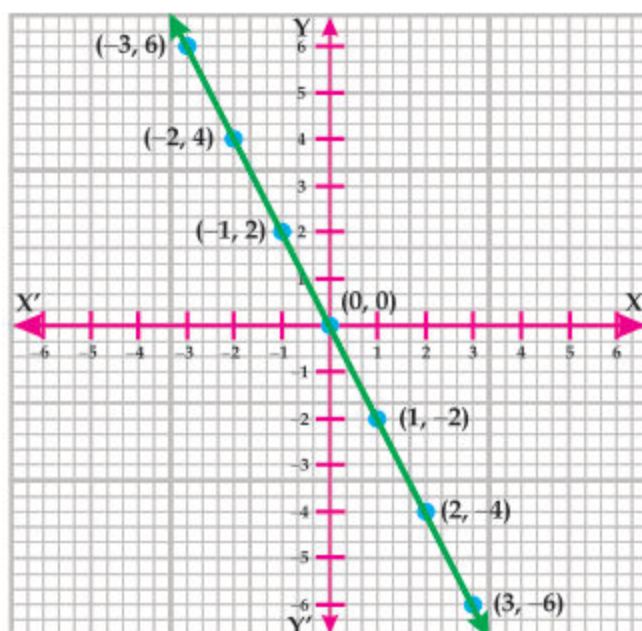
v.

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| $x$ | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| $y$ | -3  | -2  | -1  | 0   | 1   | 2   | 3   |



vi.

|     |    |    |    |   |    |    |    |
|-----|----|----|----|---|----|----|----|
| $x$ | -3 | -2 | -1 | 0 | 1  | 2  | 3  |
| $y$ | 6  | 4  | 2  | 0 | -2 | -4 | -6 |



4. i.

| Equation           | $x$ -Coordinate | $y$ -Coordinate |
|--------------------|-----------------|-----------------|
| $y = \frac{1}{2}x$ | 0               | 0               |
|                    | 4               | 2               |

ii.

| Equation           | $x$ -Coordinate | $y$ -Coordinate |
|--------------------|-----------------|-----------------|
| $x = \frac{2}{3}y$ | 1               | $\frac{3}{2}$   |
|                    | 1               | $\frac{3}{2}$   |

iii.

| Equation      | $x$ -Coordinate | $y$ -Coordinate |
|---------------|-----------------|-----------------|
| $2x + 4y = 8$ | 0               | 2               |
|               | $\frac{7}{2}$   | $\frac{1}{4}$   |

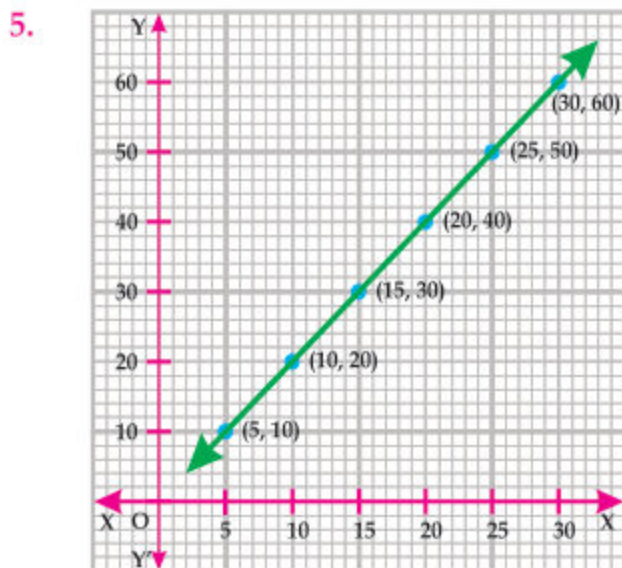
iv.

| Equation     | $x$ -Coordinate | $y$ -Coordinate |
|--------------|-----------------|-----------------|
| $2x + y = 6$ | 1               | 4               |
|              | 3               | 0               |

|    |                 |                     |                     |
|----|-----------------|---------------------|---------------------|
| v. | <b>Equation</b> | <b>X-Coordinate</b> | <b>Y-Coordinate</b> |
|    | $x - y = 2$     | 2                   | 0                   |
|    |                 | 1                   | -1                  |

|     |                 |                     |                     |
|-----|-----------------|---------------------|---------------------|
| vi. | <b>Equation</b> | <b>X-Coordinate</b> | <b>Y-Coordinate</b> |
|     | $x - 3y = 6$    | 3                   | -1                  |
|     |                 | 3                   | -1                  |



6. a. The time taken by Ayesha to ride 100 km is 5 hours.  
 b. The total distance covered by Ayesha in 3 hours is 60 km.

### Exercise 7.3

- 1.6 km
  - 4.8 km
  - 1.2 miles
  - 4.9 miles or 5 miles
- 5 acres
  - 12 acres
  - 2 hectares
  - 6 hectares
- 35.6°F
  - 35.4°F
  - 0°C
  - 2.4°C
- 90°
  - 156 Rs
  - 5 riyal
  - 2.6 riyal
- {3, -2}
  - {3, 1}
  - {3, 1}
  - {4, 7}
  - {1, 1}
  - {-3, -4}
  - {-3, 2}
  - {1, 1}
  - {8, 4}
  - {5, 1}

### Review Exercise 7

- True
  - False
  - False
  - False
- c
  - b
  - a
  - b
  - c
  - d
  - a

## Exercise 8.1

1. i.  $\{-2, -3\}$  ii.  $\left\{\frac{1}{2}, -\frac{1}{3}\right\}$  iii.  $\{6, 5\}$  iv.  $\{2, 0\}$   
 v.  $\{5, -3\}$  vi.  $\left\{\frac{8}{3}, \frac{3}{4}\right\}$  vii.  $\{8, 2\}$  viii.  $\left\{0, -\frac{8}{3}\right\}$
2. i.  $\{-3+2\sqrt{2}, -3-2\sqrt{2}\}$  ii.  $\left\{0, -\frac{10}{3}\right\}$  iii.  $\left\{\frac{4+\sqrt{13}}{3}, \frac{4-\sqrt{13}}{3}\right\}$   
 iv.  $\left\{\frac{3}{4}, -\frac{7}{6}\right\}$  v.  $\left\{4, -\frac{3}{2}\right\}$  vi.  $\left\{\frac{-2+\sqrt{6}}{2}, \frac{-2-\sqrt{6}}{2}\right\}$
3. i.  $b = -2$  ii.  $-\frac{4}{3}$

## Exercise 8.2

1. i.  $\{5, -3\}$  ii.  $\left\{\frac{3}{5}, -\frac{5}{2}\right\}$   
 iii.  $\left\{\frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right\}$  iv.  $\left\{\frac{-1+2\sqrt{7}}{3}, \frac{-1-2\sqrt{7}}{3}\right\}$   
 v.  $\left\{\frac{2+3i\sqrt{5}}{3}, \frac{2-3i\sqrt{5}}{3}\right\}$  vi.  $\left\{\frac{-3+\sqrt{41}}{4}, \frac{-3-\sqrt{41}}{4}\right\}$   
 vii.  $\left\{\frac{1+i\sqrt{5}}{3}, \frac{1-i\sqrt{5}}{3}\right\}$  viii.  $\left\{\frac{1}{2}, \frac{-1}{3}\right\}$   
 ix.  $\left\{\frac{5}{2}, 0\right\}$  x.  $\{1, -1\}$   
 xi.  $\{3\}$  xii.  $\{\sqrt{14}, -\sqrt{14}\}$

## Exercise 8.3

1.  $\{\pm 3, \pm i\}$
2.  $\{\pm 2, \pm i\}$
3.  $\left\{\pm \frac{1}{2}, \pm \frac{\sqrt{6}}{3}\right\}$
4.  $\left\{-\frac{3}{4}, -2\right\}$
5.  $\left\{\pm \frac{\sqrt{6}i}{2}, \pm \frac{2\sqrt{14}i}{7}\right\}$
6.  $\{-1, 2\}$
7.  $\{2\}$
8.  $\{-2, 1\}$
9.  $\{-1\}$
10.  $\{-2\}$
11.  $\{2, 4\}$
12.  $\left\{3, -10, \frac{-7 \pm \sqrt{79}i}{2}\right\}$
13.  $\left\{-6, 1, \frac{-5 \pm \sqrt{39}i}{2}\right\}$
14.  $\left\{0, 1, \frac{1 \pm \sqrt{57}}{2}\right\}$

## Exercise 8.4

1.  $\{4\}$
2.  $\{6\}$
3.  $\{1, 3\}$
4.  $\{2\}$
5.  $\left\{0, -\frac{3}{2}\right\}$
6.  $\{0, 3\}$

## Review Exercise

1. i. 2      ii.  $ax^2 + bx + c = 0, a \neq 0$       iii. Exponential  
iv.  $x = 2$       v.  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
2. i.  $b$       ii.  $a$       iii.  $c$       iv.  $a$       v.  $a$   
vi.  $c$       vii.  $b$       viii.  $a$       ix.  $c$       x.  $a$
3. i. False      ii. False      iii. False      iv. False  
v. False      vi. True      vii. True

## Exercise 9.1

3.  $x = 17, y = 5$

4.  $x = 5, y = 3$

## Review Exercise 9

1.  $30^\circ$

2. i. True

ii. False

iii. True

iv. False

v. False

3. i.  $\overline{DF}$

ii.  $\angle RPQ$

iii. Congruent

iv. Equilateral

v. Hypotenuse

vi.  $90^\circ$

4. i.  $b$

ii.  $c$

iii.  $b$

iv.  $c$

## Exercise 10.3

1. 8 cm

## Exercise 10.4

3. 1 cm

## Review Exercise 10

1. i. Congruent

ii. Congruent

iii. Concurrent

iv. Bisect

v. Equal

vi.  $360^\circ$

2. i.  $d$

ii.  $b$

iii.  $d$

iv.  $c$

v.  $b$

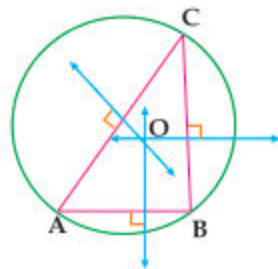
vi.  $b$

vii.  $a$

viii.  $a$

## Exercise 11.1

3. The centre of the circle passing through the three non-collinear points is on the point of intersection of right bisector of the line segment obtained by joining these non-collinear points. This point of intersection is equidistant from all three non-collinear points.



## Review Exercise 11

3. i. True

ii. True

iii. True

4. i.  $c$

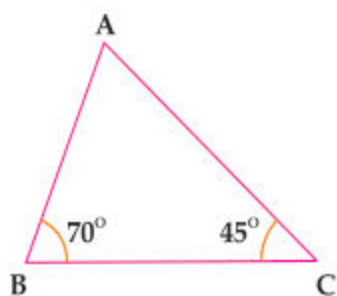
ii.  $a$

iii.  $b$

iv.  $a$

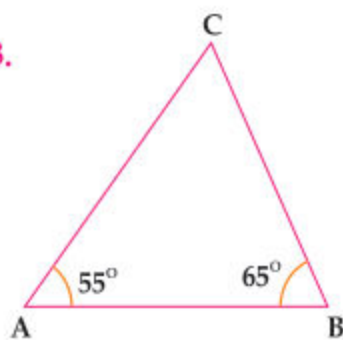
## Exercise 12.1

2.



Side AC is longest.

3.



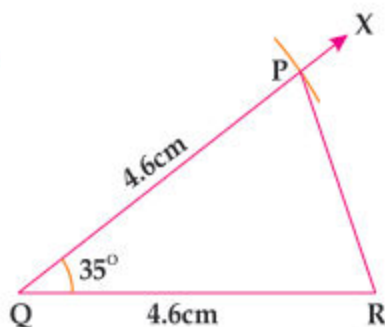
Side BC is smallest.

## Review Exercise 12

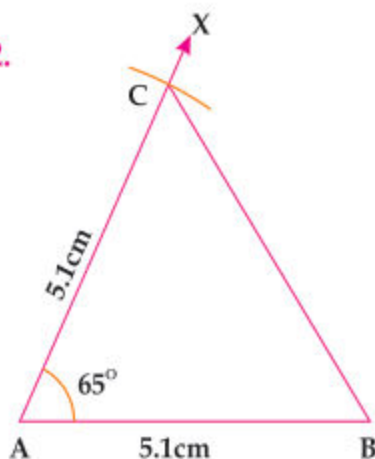
1. i. True    ii. False    iii. False    iv. False    v. True
2. i. Hypotenuse    ii. Greater    iii.  $m\overline{AB}$  is smallest
3. i. a    ii. c

## Exercise 13.1

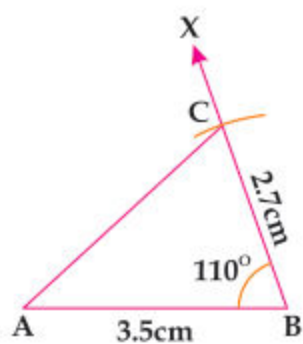
1.



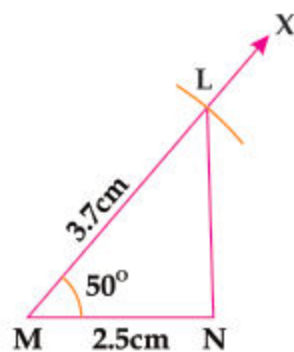
2.



3.

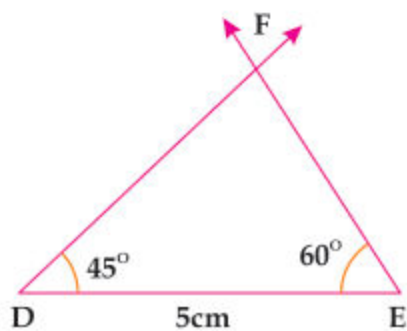


4.

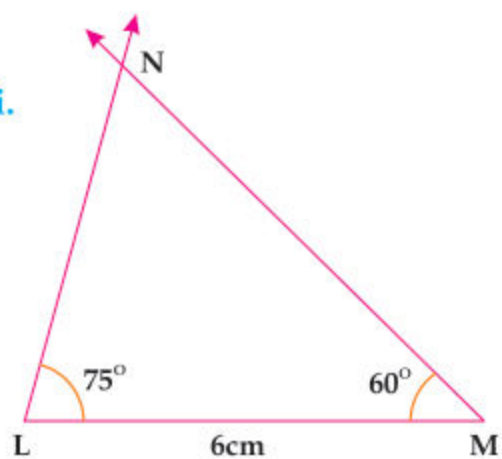


6.

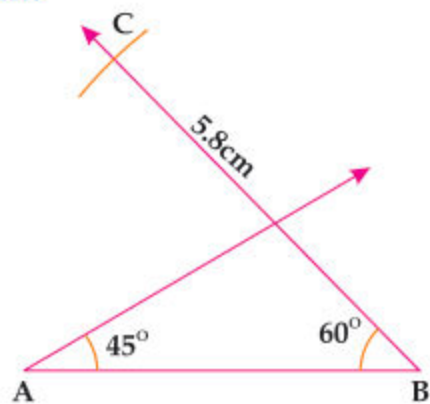
i.



ii.



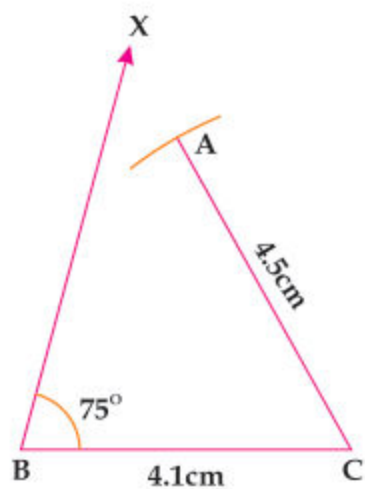
iii.



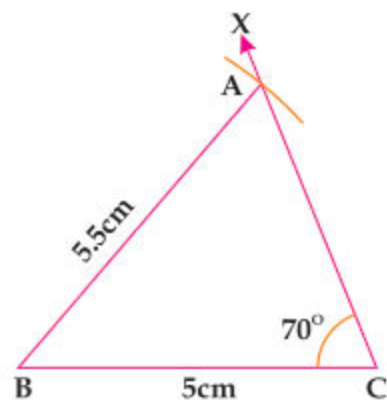


7.

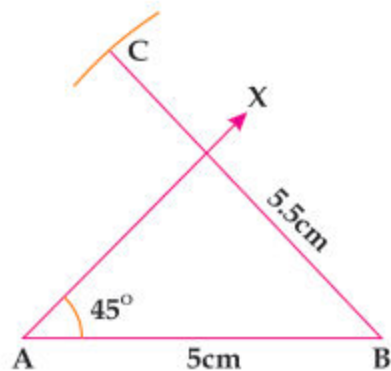
i.



ii.



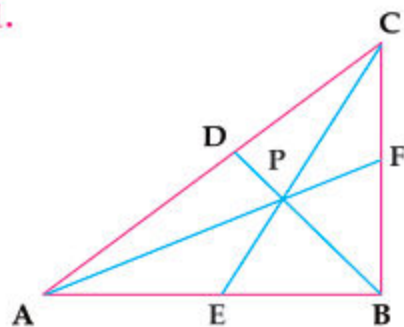
iii.



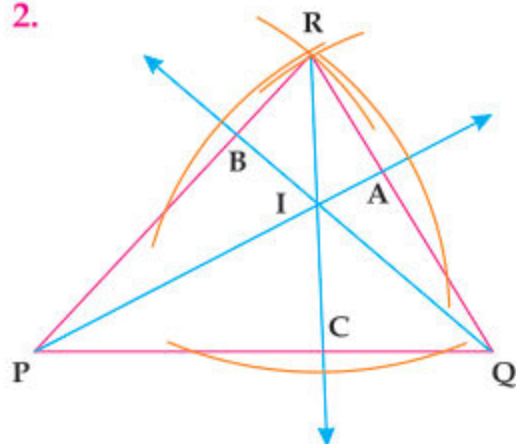
## Exercise 13.2



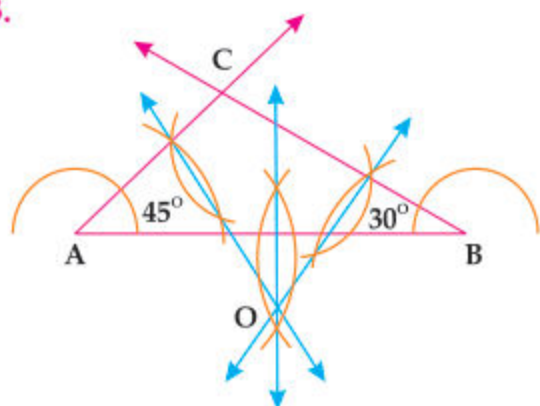
1.



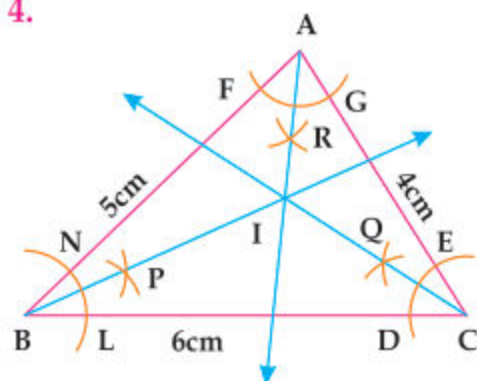
2.



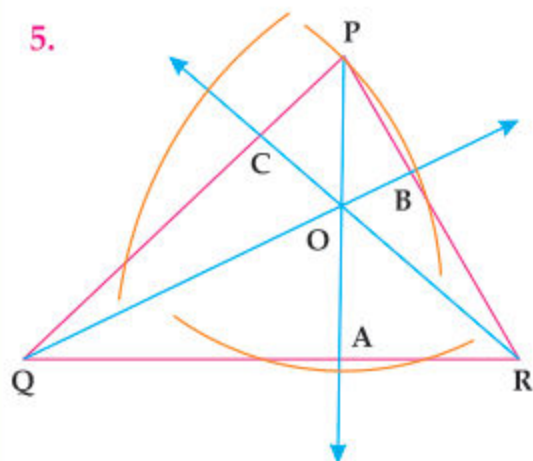
3.



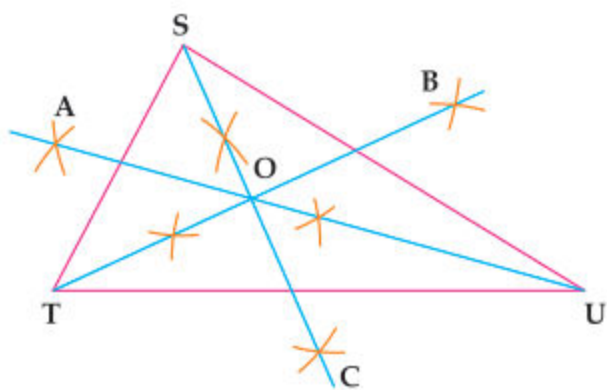
4.



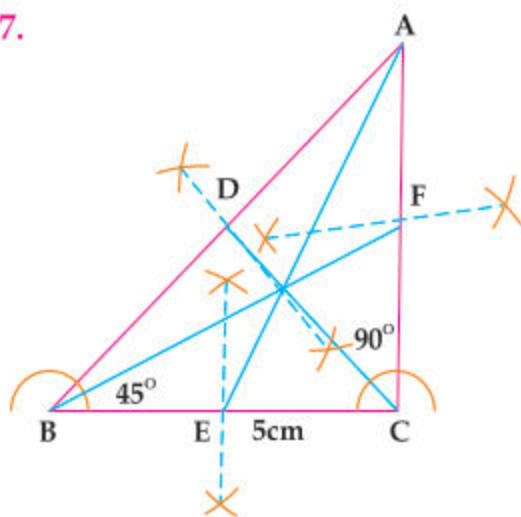
5.



6.

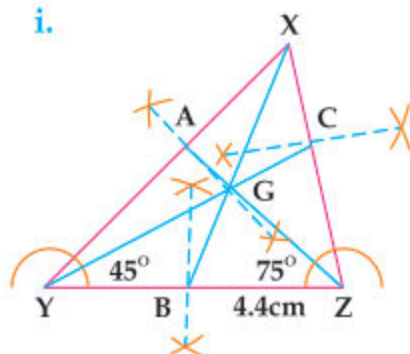


7.

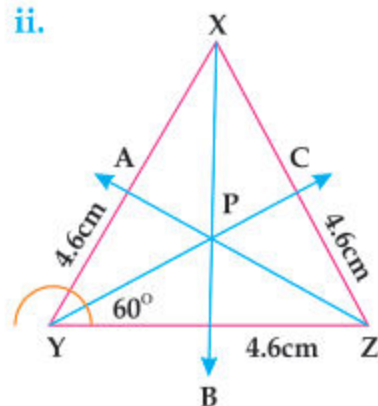


8.

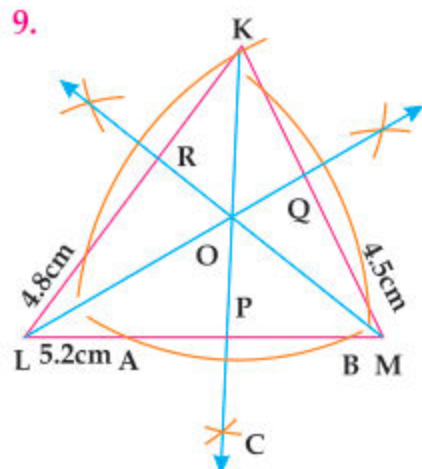
i.



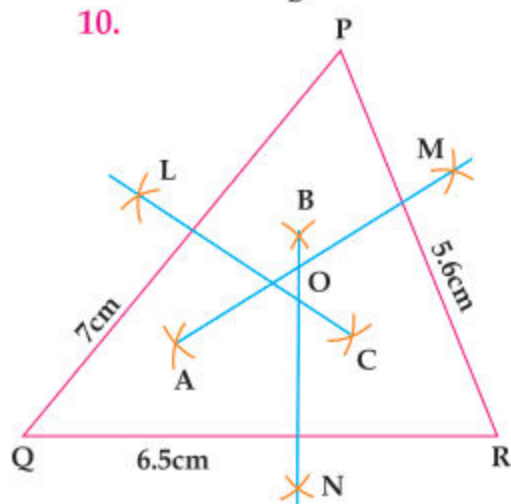
ii.



9.

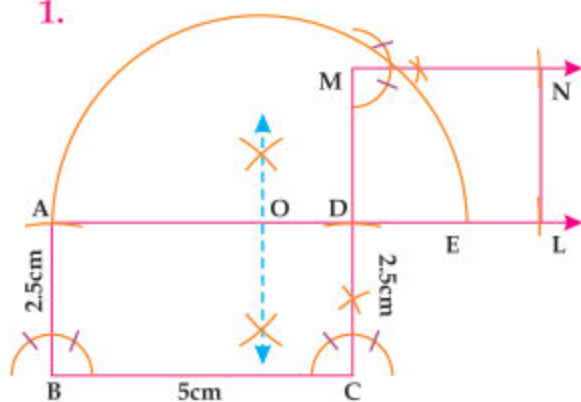


10.

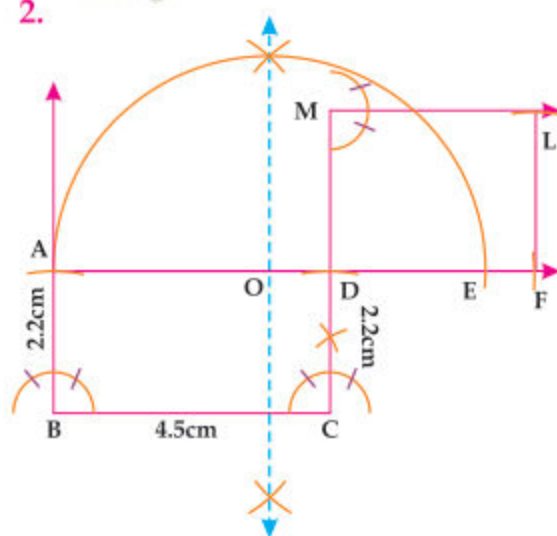


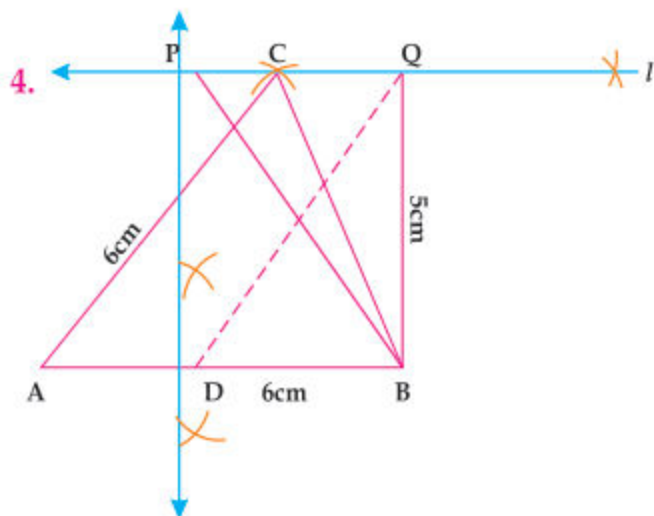
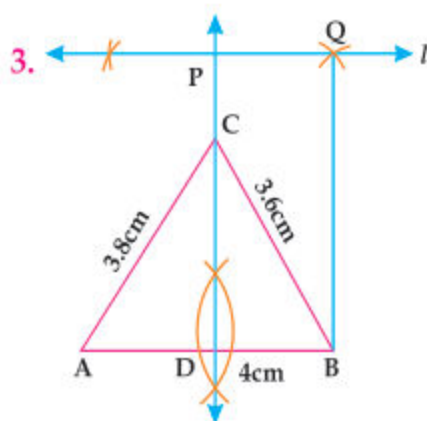
## Exercise 13.3

1.



2.





### Review Exercise 13

1. i. Hypotenuse    ii. Altitude of a triangle    iii. Median  
iv. Concurrent    v. Equal
2. i. c    ii. d    iii. c    iv. b    v. a  
vi. d    vii. c    viii. b    ix. d

### Exercise 14.1

3.  $\triangle BCD = 21\sqrt{2} \text{ cm}^2$

### Review Exercise 14

1. i. True    ii. True    iii. False    iv. True    v. False  
vi. False    vii. False    viii. True
2. i. b    ii. b    iii. c    iv. d    v. d    vi. b

## Exercise 15.1

- Side  $m\overline{AB} = 3\sqrt{7} \text{ cm}$ , Area  $= \frac{9\sqrt{3}}{2} \text{ cm}^2$
  - Length of side  $m\overline{AB} = 4\sqrt{7} \text{ cm}$ , Area  $= 8\sqrt{3} \text{ cm}^2$
- Length of side  $\overline{AC} = \sqrt{116} \text{ cm}$ , Area  $= 12 \text{ cm}^2$
- Length of side  $AC = \sqrt{232} \text{ cm}$ , Area  $= 24 \text{ cm}^2$

## Exercise 15.2

- $m\overline{BC} = 46 \text{ cm}$
- Right triangle
- length of median  $\sqrt{\frac{7}{2}} \text{ cm}$

## Review Exercise 15

- $(m\overline{BC})^2$
  - Isosceles
  - $(m\overline{AC})^2$
  - Triangle
- $m\overline{AB} = 8 \text{ cm}$

## Exercise 16.1

- $\sqrt{101}$
  - 1
  - $2\sqrt{2}$
  - $2\sqrt{2}$
- 5
  - $\sqrt{117}$
  - 5
  - $\sqrt{10}$
- $P = 12$

## Exercise 16.2

- Points A, B and C forms isosceles triangle.
- Point A, B and C form right angled triangle.
- $k = 1 \pm 3\sqrt{3}$
- Because squares have equal sides and these points determine square.

## Exercise 16.3

1. i.  $(-1, 7)$       ii.  $\left(1, \frac{1}{2}\right)$       iii.  $(-4, 3)$       iv.  $(\sqrt{3}, 2\sqrt{3})$
2.  $(-1, 1)$       3.  $B = (2, 2)$       4. center  $(4, 5)$   
radius  $= 5\sqrt{2}m$

## Review Exercise 16

1. i. True      ii. False      iii. False      iv. False      v. True  
vi. True      vii. True      viii. True      ix. False
2. i.  $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$       ii. Line      iii.  $y < 0$
3. i.  $c$       ii.  $a$       iii.  $d$       iv.  $d$